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Neural PK-PD Modeling with ODE: Multi-Task Molecular Property Prediction and Decision-Oriented Simulation

Sushree Smaranika PRADHAN
student record book number 123456

thesis supervisor
title/degree Name xyzzz

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Abstract

This thesis presents an end-to-end machine learning framework for early drug development that combines molecular property prediction with mechanistic pharmacokinetic-pharmacodynamic (PK-PD) simulation. The workflow integrates heterogeneous public datasets, applies feature engineering and data-quality controls, and trains multi-task neural models to predict binding affinity, hERG-related cardiotoxicity, Caco-2 permeability, and clearance.

The central contribution is a physics-informed decision pipeline in which predicted molecular endpoints are coupled with a Neural ODE-based PK-PD simulator to support translational analysis. The experimental sequence begins with a notebook-aligned molecular-representation upgrade for the TDC-derived tasks, then proceeds to real ChEMBL binding-data integration, hERG-focused predictive refinement, probability calibration for safety classifiers, structure-aware joint GNN+MLP fusion modeling, mechanistic PK-PD simulation, Pareto efficacy-safety trade-off analysis, Monte Carlo robustness assessment, and in-silico clinical trial simulation. A RapidDock-inspired, distance-biased transformer branch is included as an exploratory prototype for future structure-aware extension.

The results show that predictive quality improves materially after real-data integration and fusion modeling, that post-hoc calibration is necessary before classifier outputs can be used as risk variables, and that the resulting PK-PD framework supports clinically motivated trade-off and trial-level analyses. The thesis concludes by discussing practical limitations, reproducibility considerations, and future directions for external validation and deployment-oriented decision support.

Keywords: drug discovery, PK-PD, Neural ODE, multi-task learning, graph neural networks, calibration, dose optimization

Streszczenie

Niniejsza praca przedstawia kompleksowe podejście z zakresu uczenia maszynowego do wspomagania wczesnego rozwoju leków, łączące predykcję właściwości molekularnych z mechanistyczną symulacją farmakokinetyczno-farmakodynamiczną (PK-PD). Zaproponowany przepływ pracy integruje heterogeniczne publiczne zbiory danych, obejmuje inżynierię cech i kontrolę jakości danych oraz wykorzystuje wielozadaniowe modele neuronowe do przewidywania powinowactwa wiązania, kardiotoksyczności związanej z hERG, przepuszczalności Caco-2 oraz klirensu.

Główny wkład pracy stanowi fizycznie ugruntowany pipeline decyzyjny, w którym przewidywane molekularne punkty końcowe są sprzęgnięte z symulatorem PK-PD opartym na Neural ODE w celu prowadzenia analiz translacyjnych. Praca realizuje uporządkowaną sekwencję eksperymentalną obejmującą przejściowe ulepszenie reprezentacji molekularnej dla zadań pochodzących z TDC, późniejszą integrację rzeczywistych danych wiązania z ChEMBL, ukierunkowane doskonalenie predykcji hERG, kalibrację prawdopodobieństw dla klasyfikatorów bezpieczeństwa, modelowanie strukturalne z użyciem wspólnej architektury fuzji GNN+MLP, mechanistyczną symulację PK-PD, analizę kompromisów skuteczność–bezpieczeństwo w ujęciu Pareto, ocenę odporności metodą Monte Carlo oraz symulację badań klinicznych in silico. Dodatkowo uwzględniono eksploracyjną gałąź inspirowaną RapidDock, wykorzystującą transformer z biasem odległościowym jako prototyp przyszłego modelowania uwzględniającego strukturę.

Wyniki pokazują, że jakość predykcji istotnie wzrasta po integracji danych rzeczywistych i zastosowaniu modelu fuzji, że kalibracja post-hoc jest konieczna przed wykorzystaniem wyjść klasyfikacyjnych jako zmiennych ryzyka, oraz że otrzymany framework PK-PD wspiera klinicznie motywowane analizy kompromisów i symulacje na poziomie badań klinicznych. Pracę zamyka omówienie praktycznych ograniczeń, zagadnień reprodukowalności oraz przyszłych kierunków badań ukierunkowanych na walidację zewnętrzną i wdrożeniowe wspomaganie decyzji.

Słowa kluczowe: odkrywanie leków, PK-PD, Neural ODE, uczenie wielozadaniowe, grafowe sieci neuronowe, kalibracja, optymalizacja dawkowania

Contents

Abstract **2**

Streszczenie **3**

0.1 Objectives and Contributions 8

0.2 Thesis Organization 8

1 Background and Theoretical Foundations **10**

1.1 ADME-Tox Endpoints in Early Drug Discovery 10

1.2 Multi-Task Learning 10

1.3 Graph Neural Networks for Molecular Modeling 10

1.4 PK-PD Modeling with ODEs 10

1.5 Calibration and Decision Reliability 11

2 Related Work and Research Gap **14**

2.1 Molecular Representations for Property Prediction 14

2.2 Multi-Task Learning and ADMET Modeling 14

2.3 Mechanistic PK-PD Modeling and Neural ODEs 15

2.4 Calibration and Decision-Oriented Modeling 15

2.5 Research Gap 16

3 Problem Formulation, Objectives, and Scope **18**

3.1 Task Definitions 18

3.2 Evaluation Metrics 19

3.3 Scope and Constraints 21

4 Data Sources, Curation, and Preprocessing **22**

4.1 Chapter Overview 22

4.2 Data Sources and Scientific Role 22

4.2.1 PubChem: Assays and Exemplar Compounds 22

4.2.2 PK-DB: Pharmacokinetic Studies and Parameters 23
4.2.3 TDC: Standardized ADMET Benchmarks 23
4.2.4 ChEMBL: Binding Affinity Data 23
4.2.5 ToxCast/Tox21-Style Toxicity Reference Data 24
4.3 Data Curation and Harmonization 25
4.4 Summary of Source Roles in the Framework 25
4.5 Data Validation and Quality Assurance 26
4.5.1 Validation Checks in the Pipeline 26
4.5.2 Critical Files for Downstream Modeling 26
4.6 Exploratory Data Analysis and Data Quality Assessment 27
4.6.1 Dataset Loading and Composition 27
4.6.2 ChEMBL Binding Affinity Analysis 28
4.6.3 ToxCast Toxicity Assessment 28
4.6.4 TDC ADMET Properties Analysis 30
4.6.5 Pharmacokinetic Studies (PK-DB) 31
4.6.6 Data Quality and Integrity Assessment 31
4.6.7 Multi-Task Learning Dataset Assembly 32
4.6.8 Feature Preprocessing and Normalization 33
4.7 Feature Engineering 34
4.7.1 Molecular Representations and Feature Space Design 34
4.7.2 Task-Specific Feature Sets and Multi-Task Integration 35
4.7.3 Docking-Derived Bridge Signal (RapidDock Prototype) 36
4.7.4 Feature Storage and Reusability 36
4.7.5 Hyperparameter Configuration 36
4.7.6 Data Quality Gate 38
4.8 Quality Control and Split Strategy 39
4.8.1 Quality Control Procedures 39
4.8.2 Train/Validation/Test Split Strategy 40
4.8.3 Metadata and Provenance Tracking 40

5 Methodology and Model Architecture 42
5.1 Exploratory Data Analysis Methods 42
5.1.1 Descriptive Statistical Analysis 42
5.1.2 Distribution Visualization 42
5.1.3 Data Quality Checks 43

5.2 Multi-Task Learning Architecture 43
5.2.1 Model Design Notebook Data Interface 43
5.2.2 Shared Encoder Design 44
5.2.3 Task-Specific Heads 44
5.2.4 Multi-Task Objective 45
5.3 Full Multi-Task Neural ODE Architecture 45
5.3.1 Objectives and Target Metrics 45
5.3.2 RapidDock Bridge Feature Augmentation 45
5.3.3 Layer-by-Layer Architecture 46
5.3.4 Training Loop Design 47
5.3.5 Loss Function 47
5.3.6 Validation and Threshold Selection 48
5.4 Fine-Tuning, Calibration, and Leakage Mitigation 48
5.4.1 Task-Specific Fine-Tuning (Variants 1 and 2) 48
5.4.2 Threshold Calibration 49
5.4.3 Split-Leakage Mitigation 50
5.5 Fingerprint Upgrade: From Synthetic to Real SMILES 50
5.5.1 Motivation for Fingerprint Upgrade 50
5.5.2 2048-Bit Real-SMILES Upgrade 51
5.5.3 Current Notebook-Aligned Representation Upgrade 51
5.6 Fine-Tuning Strategy 52
5.6.1 hERG AUROC Push (hidden_dim=256 + pos_weight sweep) 52
5.7 Graph Encoder and Fusion Architecture 52
5.7.1 Pure-PyTorch GNN Molecular Encoder 52
5.7.2 End-to-End Joint GNN and MLP Fusion 53
5.8 Neural ODE PK-PD Integration 53
5.8.1 Mechanistic Simulation Design 53
5.9 Probability Calibration 54
5.9.1 Temperature Scaling and Platt Scaling 54
5.10 RapidDock-Inspired Alternative Branch 55
5.10.1 Input Representation and Data Pipeline 55
5.10.2 Distance-Biased Multi-Head Attention 56
5.10.3 Architecture: RapidDockPrototype 56
5.10.4 Symmetry-Aware Loss 57

5.10.5 Ligand Coordinate Reconstruction (L-BFGS) 57

6 Experimental Setup and Protocol 59

6.1 Environment and Reproducibility 59

6.2 Training Protocol 60

6.3 Experimental Sequence 60

6.4 RapidDock Prototype Protocol 63

7 Results 64

7.1 Chapter Overview 64

7.2 Baseline Multi-Task Predictive Performance 64

7.3 Transitional Molecular Representation Upgrade 65

7.4 Real Binding Data Integration 66

7.5 hERG Refinement and Final Fusion Model 66

7.6 Neural ODE PK-PD Integration 68

7.7 Probability Calibration and Robustness 69

7.8 Pareto Efficacy-Safety Trade-Off Analysis 70

7.9 Virtual Clinical Trial Simulation 71

7.10 Exploratory Extensions 72

7.11 Summary of Key Findings 72

8 Discussion 73

8.1 Interpretation of Key Findings 73

8.1.1 Structural Representation Is a Necessary Starting Point 73

8.1.2 Real Binding Data Provides the First Major Task-Specific Gain 74

8.1.3 Predictive Refinement and Calibration Must Precede Translation 74

8.1.4 Fusion Modeling Benefits from Complementary Structural Views 74

8.2 Mechanistic PK-PD Interpretation 75

8.3 Pareto, Monte Carlo, and Virtual Trial Implications 75

8.4 Limitations 76

8.5 Comparison with Related Work 77

9 Conclusion and Future Work 78

Appendix 79

References 94

The practical motivation is to move from isolated endpoint prediction to coherent therapeutic assessment under uncertainty.

0.1. Objectives and Contributions

The thesis has four principal objectives:

1. Build a high-quality, leakage-controlled multi-task dataset from heterogeneous public sources.
2. Develop predictive models for binding affinity, hERG risk, Caco-2 permeability, and clearance, with explicit evaluation of real-data integration and structure-aware fusion modeling.
3. Integrate calibrated model outputs with Neural ODE-based PK-PD simulation for mechanistic and translational analysis.
4. Evaluate decision-oriented behavior through Pareto trade-off analysis, Monte Carlo robustness assessment, and virtual clinical trial simulation.

The key contributions are:

1. A full experimental pipeline spanning data curation, feature engineering, predictive modeling, calibration, mechanistic simulation, and translational decision analysis.
2. A thesis-essential modeling sequence showing that the notebook-aligned TDC-focused representation upgrade is only a transitional reference stage, while later real binding-data integration, hERG-focused refinement, and joint GNN+MLP fusion produce the strongest predictive basis for downstream PK-PD analysis.
3. A decision-support layer combining Pareto trade-off analysis, Monte Carlo validation, and virtual clinical trial simulation on top of calibrated predictive outputs.
4. An exploratory RapidDock-inspired prototype branch with documented protocol and qualitative feasibility assessment.

0.2. Thesis Organization

Table 1 summarizes the organization of the thesis.

Chapter	**Content Summary**
Chapter 2	Theoretical background for ADME-Tox modeling, multi-task learning, graph neural networks, PK-PD modeling, and calibration.
Chapter 3	Related work and the research gap addressed by the thesis.
Chapter 4	Problem formulation, task definitions, evaluation metrics, objectives, and scope.
Chapter 5	Data sources, curation, harmonization, feature engineering, and quality-control strategy.
Chapter 6	Methodology, model architectures, calibration methods, Neural ODE integration, and the RapidDock-inspired branch.
Chapter 7	Experimental environment, training protocol, and the thesis-essential experimental sequence.
Chapter 8	Empirical results across real-data integration, predictive refinement, calibration, PK-PD simulation, Pareto analysis, Monte Carlo validation, and virtual clinical evaluation.
Chapter 9	Discussion of findings, interpretation, limitations, and comparison with related work.
Chapter 10	Conclusions and future research directions.

Table 1: Organization of the thesis.

1. Background and Theoretical Foundations

1.1. ADME-Tox Endpoints in Early Drug Discovery

Absorption, distribution, metabolism, excretion, and toxicity (ADME-Tox) endpoints are central to candidate selection. In this thesis, the following endpoints are modeled:

1. binding affinity (regression),
2. hERG-related cardiotoxicity risk (classification),
3. Caco-2 permeability proxy (classification),
4. clearance (regression).

1.2. Multi-Task Learning

Multi-task learning uses shared representations across related tasks to improve generalization, especially under partial labels [7]. A shared encoder with task-specific heads is used in this work to model common molecular factors while preserving task-specific outputs.

1.3. Graph Neural Networks for Molecular Modeling

Molecules are naturally represented as graphs, where atoms are nodes and bonds are edges. Message Passing Neural Networks (MPNNs) [2] aggregate local structural information into latent embeddings that can be used for property prediction [6].

1.4. PK-PD Modeling with ODEs

PK-PD modeling connects drug concentration in the body to its biological effect in two linked stages.

Pharmacokinetic (PK) stage describes how drug concentration evolves over time using a two-compartment ODE system:

$$\frac{dC_c}{dt} = -\frac{CL}{V_c}C_c - k_{12}C_c + k_{21}C_t + u(t), \tag{1.1}$$

$$\frac{dC_t}{dt} = k_{12}C_c - k_{21}C_t, \tag{1.2}$$

where C_c is drug concentration in the *central compartment* (blood/plasma), C_t is drug concentration in the *peripheral compartment* (tissues), k_{12} and k_{21} are inter-compartmental transfer rate constants, CL/V_c is the elimination rate constant (clearance divided by volume), and $u(t)$ is the dosing function (e.g. IV bolus).

Pharmacodynamic (PD) stage links drug concentration to biological effect via the Hill–Emax model with effect-site linkage:

$$E(t) = E_{\max} \cdot \frac{C_c(t)^\gamma}{EC_{50}^\gamma + C_c(t)^\gamma}, \tag{1.3}$$

where $E_{\max}$ is the maximum achievable pharmacological effect, EC_{50} is the concentration producing 50% of maximal effect (derived from predicted binding affinity as $EC_{50} = 10^{-\text{pChEMBL}}$), and γ is the Hill coefficient controlling the steepness of the response curve.

The predicted molecular endpoints are mapped to ODE parameters as follows:

Predicted property	**ODE/PK-PD role**
Hepatic clearance (regression)	Elimination rate constant $k_e = CL/V_c$
Caco-2 permeability (probability)	Oral absorption rate k_a
Binding affinity pChEMBL (regression)	Pharmacodynamic potency EC_{50}
hERG probability (classification)	Cardiac safety flag

Table 1.1: Mapping of predicted molecular properties to ODE pharmacokinetic/pharmacodynamic parameters.

The ODEs are solved numerically using `torchdiffeq.odeint` (Dormand–Prince RK45 solver), which makes the entire pipeline differentiable and enables physics-informed training through the ODE solution if required.

1.5. Calibration and Decision Reliability

For classification tasks, probability calibration is required when outputs are used in downstream risk computations. This thesis evaluates Expected Calibration Error (ECE) and applies temperature scaling and Platt scaling [10].

The Problem: Discrimination vs. Reliability

A model can rank compounds correctly (high AUROC) while still assigning systematically wrong probability values. For example, predicting a 95% hERG inhibition risk when the true empirical rate is 60% makes the model *overconfident*. Since predicted probabilities are passed directly into downstream PK-PD simulation—as the cardiac safety flag and the Caco-2-derived oral absorption rate k_a—this bias propagates into dosing decisions and operating characteristics. Calibration corrects this by aligning predicted confidence with observed accuracy.

Expected Calibration Error (ECE)

The primary calibration metric used in this thesis partitions predicted probabilities into $M = 10$ equal-width bins and compares the average predicted confidence to the actual accuracy within each bin:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{1.4}$$

where B_m is the set of samples in bin m, $\mathrm{acc}(B_m)$ is the fraction of positive labels in that bin, and $\mathrm{conf}(B_m)$ is the mean predicted probability. A perfectly calibrated model achieves $\mathrm{ECE} = 0$.

Calibration Methods

Two post-hoc calibration techniques are evaluated:

1. **Temperature Scaling** [10]: divides the raw logit z by a scalar temperature $T > 0$ before applying the sigmoid:

   $$\hat{p} = \sigma(z/T), \quad T > 0. \tag{1.5}$$

   $T > 1$ softens overconfident predictions; $T < 1$ sharpens underconfident ones. The temperature is fitted by minimising negative log-likelihood on the validation set.

2. **Platt Scaling**: applies an affine transformation to the logit:

   $$\hat{p} = \sigma(a \cdot z + b), \tag{1.6}$$

   where parameters (a, b) are learned by logistic regression on validation-set logits and labels.

Results

Post-hoc calibration substantially reduced ECE for both classification tasks:

Calibrated probabilities are used directly in all downstream Neural ODE simulation, Pareto optimisation, and in-silico trial analyses. Uncalibrated outputs would introduce systematic bias into every PK-PD simulation and dose recommendation produced by the framework.

Task	**ECE before**	**ECE after**
hERG (temperature scaling)	0.1727	**0.0058**
Caco-2 (Platt scaling)	0.3583	**0.0448**

Table 1.2: Expected Calibration Error before and after post-hoc probability calibration.

2. Related Work and Research Gap

2.1. Molecular Representations for Property Prediction

Recent work in drug discovery has shown that deep learning can achieve strong predictive performance for molecular property estimation when appropriate structural representations are used [12]. Earlier cheminformatics pipelines relied heavily on fixed descriptors and fingerprints, which remain attractive because they provide compact summaries of local chemical environments and can be used efficiently across large compound libraries [5]. Later work expanded this representation space through graph neural networks, which operate directly on atom–bond structure rather than on hand-crafted vectors [6, 2], and through attention-based architectures derived from transformers [3]. Benchmark efforts such as MoleculeNet further reinforced the importance of representation design by enabling consistent comparison of descriptors, graph models, and learned molecular embeddings across multiple tasks [14].

These studies establish a clear methodological trend: as molecular representation becomes more expressive, predictive performance and transferability tend to improve, particularly when tasks depend on subtle structure–activity relationships. This observation is directly relevant to the present thesis because it motivates the comparison between fingerprint-based baselines, graph encoders, and structure-aware extensions.

2.2. Multi-Task Learning and ADMET Modeling

Within ADMET and bioactivity prediction, much of the literature has focused on endpoint-specific problems such as oral absorption, toxicity, and binding prediction [15, 16, 11]. At the same time, multi-task learning has emerged as an important strategy for exploiting related pharmacological endpoints within a shared model. In particular, massively multi-task neural networks demonstrated that parameter sharing can improve predictive performance by transferring information across assays and targets that are only partially labeled [13]. Public resources such as ChEMBL have further enabled large-scale supervised learning on molecular activity data [17], while benchmark-oriented studies have encouraged standardized model comparison across diverse molecular tasks [14].

Despite this progress, a substantial portion of the ADMET literature still evaluates models primarily in terms of isolated predictive metrics for individual endpoints or small collections of related tasks. From a pharmaceutical perspective, this is limiting, because practical candidate evaluation depends on the joint interpretation of potency, safety, permeability, and clearance rather than on any one property in isolation. The present thesis therefore builds on the multi-task learning tradition, but places stronger emphasis on cross-endpoint integration and downstream use.

2.3. Mechanistic PK-PD Modeling and Neural ODEs

Pharmacokinetic-pharmacodynamic research has long provided mechanistic tools for relating dose, concentration, and biological response through systems of ordinary differential equations [4]. Such models are valuable because they translate static parameters into clinically interpretable trajectories, including concentration–time and effect–time profiles. Neural Ordinary Differential Equations generalize this perspective by offering a differentiable framework for modeling continuous dynamics within modern deep-learning systems [1]. This creates a natural opportunity to connect molecular prediction to dynamic simulation rather than treating endpoint estimation as the final output.

However, these two literatures often remain separated in practice. PK-PD studies are typically centered on mechanistic or clinical modeling, while molecular machine-learning studies usually stop at endpoint prediction and do not propagate predicted properties into a dynamic simulation layer. This separation leaves a gap between model development and therapeutic interpretation.

2.4. Calibration and Decision-Oriented Modeling

Another limitation in the predictive modeling literature is that probability quality is often treated as secondary to ranking or classification accuracy. For decision-oriented tasks, however, probability calibration is essential, because overconfident or underconfident outputs can distort downstream risk assessment and resource allocation. This issue has been emphasized in the calibration literature, where post-hoc methods such as temperature scaling provide effective correction of neural-network confidence estimates [10]. In the context of drug discovery, this distinction is especially important when classification outputs are reused as risk variables or surrogate parameters in subsequent modeling stages.

For the present thesis, calibration is therefore not merely a reporting refinement. It is a methodological requirement because hERG and Caco-2 probabilities are carried forward into PK-PD and decision analyses. This shifts the role of calibration from an optional machine-learning diagnostic to a necessary component of simulation reliability.

2.5. Research Gap

The relevant literature strands and their limitations are summarized in Table 2.1. Existing studies collectively provide strong foundations for molecular representation learning, multi-task prediction, benchmark evaluation, mechanistic PK-PD modeling, and calibration. Yet they typically emphasize only one or two of these components at a time. As a result, there remains limited support for a single reproducible workflow that begins with heterogeneous molecular datasets and ends with decision-oriented PK-PD simulation.

The research gap addressed by this thesis is therefore not merely the absence of another molecular predictor, but the lack of a reproducible end-to-end framework that connects multi-task molecular property prediction to mechanistic and decision-oriented simulation in a single workflow. The contribution of the thesis is to integrate heterogeneous public datasets, shared molecular representations, calibrated classification outputs, Neural ODE-based PK-PD modeling, and population-level decision analyses within one coherent pipeline. In this sense, the work aims to bridge the methodological gap between benchmark-oriented molecular prediction and simulation-based therapeutic decision support.

Literature strand	**Typical strength**	**Common limitation relative to this thesis**
Descriptor and fingerprint models	Efficient molecular encoding and competitive baseline prediction	Limited ability to model higher-order structural interactions or dynamic downstream consequences
Graph and attention-based molecular models	Richer structure-aware learning for molecular property prediction	Often evaluated as standalone predictors rather than as components of an end-to-end decision framework
Multi-task ADMET and bioactivity learning	Shared representations across related endpoints and assays	Frequently optimized for benchmark prediction metrics without mechanistic simulation or calibrated decision use
Mechanistic PK-PD modeling	Interpretable concentration–time and effect–time simulation	Usually parameterized from experimental or clinical quantities rather than from learned molecular predictions
Calibration methods	Improved reliability of predicted probabilities	Commonly treated as a final evaluation step instead of an integrated prerequisite for downstream simulation

Table 2.1: Summary of the main literature strands relevant to the thesis and the gap addressed by their incomplete integration.

3. Problem Formulation, Objectives, and Scope

3.1. Task Definitions

The entire framework is formulated as a *multi-task learning* problem. Let x denote the molecular input feature vector for a drug candidate, comprising 2,050 dimensions: 2,048 Morgan fingerprint bits (radius 2) plus molecular weight (MW) and octanol-water partition coefficient (logP). The joint target vector spans four pharmacologically important endpoints:

$$y = [y_{\text{bind}},\ y_{\text{hERG}},\ y_{\text{Caco-2}},\ y_{\text{CL}}] \,. \tag{3.1}$$

Tasks are defined as:

1. $f_{\text{bind}}(x)$: **Regression of binding affinity** (pIC50, range 3–11). Predicts how strongly the drug binds to its pharmacological target. Higher pIC50 indicates greater potency. Values are sourced from ChEMBL across 8 mechanistically diverse protein targets (kinases, GPCRs, nuclear receptors).

2. $f_{\text{hERG}}(x)$: **Classification probability of hERG channel inhibition**. Predicts the probability that the drug blocks the hERG cardiac potassium ion channel, which is a primary cause of drug-induced QT prolongation and potentially fatal arrhythmias. A high probability flags a cardiotoxicity liability.

3. $f_{\text{Caco-2}}(x)$: **Classification probability for intestinal permeability class**. Predicts the probability of high Caco-2 cell permeability (favourable oral absorption). High permeability correlates with good bioavailability; low permeability indicates absorption-limited compounds.

4. $f_{\text{CL}}(x)$: **Regression of hepatic clearance** (mL/min/kg). Predicts the rate at which the liver metabolises the drug, which controls how long it remains in the body. High clearance compounds require frequent dosing; low clearance compounds may accumulate.

Rationale for Multi-Task Learning

A single shared encoder learns molecular representations common to all four endpoints simultaneously, since properties such as molecular weight, ring systems, and functional groups influence multiple ADMET properties. Task-specific heads then branch from this shared bottleneck for individual predictions. This promotes positive transfer learning under uneven task sizes, allowing smaller endpoints such as Caco-2 (900 samples in the baseline harmonized matrix) and hERG (655 samples) to benefit from structural representations learned jointly with binding affinity (2,000 samples) and clearance (1,213 samples).

Connection to PK-PD Simulation

The four task outputs are directly mapped to pharmacokinetic and pharmacodynamic ODE parameters:

Task output	**ODE role**	**Parameter**
$f_{\text{bind}}(x)$ – pChEMBL	Pharmacodynamic potency	$EC_{50} = 10^{-\text{pChEMBL}}$
$f_{\text{Caco-2}}(x)$ – probability	Oral absorption rate	k_a
$f_{\text{CL}}(x)$ – clearance	Elimination rate constant	$k_e = CL/V_c$
$f_{\text{hERG}}(x)$ – probability	Cardiac safety flag	Safety threshold

Table 3.1: Mapping of multi-task model outputs to Neural ODE pharmacokinetic/pharmacodynamic parameters.

3.2. Evaluation Metrics

Regression tasks are evaluated with R^2 and RMSE. Classification tasks are evaluated with AUROC, F1, and accuracy, with calibrated probability quality measured by ECE.

Regression Metrics

1. **Coefficient of Determination (R^2):** measures the fraction of variance in the target explained by the model:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{3.2}$$

   $R^2 = 1$ indicates perfect prediction; $R^2 = 0$ means the model performs no better than predicting the mean; negative values indicate the model is worse than the mean baseline.

2. **Root Mean Squared Error (RMSE)**: measures the average prediction error in the original units, penalising large errors more than small ones:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{3.3}$$

   Lower RMSE is better. RMSE is reported in normalised units for clearance and in pIC50 units for binding affinity.

Classification Metrics

1. **AUROC** (Area Under the Receiver Operating Characteristic Curve): measures ranking quality across all classification thresholds. A value of 0.5 corresponds to random discrimination; 1.0 is perfect. AUROC is the primary classification metric because it is threshold-independent and robust to class imbalance, which is present in both hERG and Caco-2 classification tasks. In the current Phase 2 harmonized matrix, the Caco-2 task is explicitly median-binarized to an approximately balanced 50% positive rate.

2. **F1 Score**: harmonic mean of precision and recall:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.4}$$

   F1 is particularly important for safety-critical endpoints such as hERG, where false negatives (missed cardiotoxic compounds) carry direct clinical risk.

3. **Accuracy**: fraction of correctly classified samples. Used alongside F1 as a guardrail during threshold calibration (minimum accuracy $\geq$ 0.50 required before locking a production threshold).

4. **Expected Calibration Error (ECE)**: measures whether predicted probabilities reflect true empirical frequencies (see Section 2.5). ECE is critical because calibrated probabilities are passed directly into downstream Neural ODE simulation as absorption and safety parameters.

Research Targets and Achieved Results

The following performance thresholds were set at the start of the project and evaluated against the final model:

All four targets were first met simultaneously at model_15 (hERG AUROC push, Section 8.4.3); binding $R^2 \geq 0.40$ was further improved to 0.550 by the joint GNN+MLP fusion model.

Task	Metric	Target	Achieved
hERG inhibition	AUROC	≥ 0.80	**0.821**
Caco-2 permeability	AUROC	≥ 0.75	**0.864**
Hepatic clearance	R^2	≥ 0.20	**0.348**
Binding affinity	R^2	≥ 0.40	**0.550**

Table 3.2: Research performance targets and final achieved metrics (model_15 / joint GNN+MLP fusion).

3.3. Scope and Constraints

The thesis focuses on data-driven preclinical decision support. It does not claim prospective clinical validation. External validation and deployment-grade clinical calibration are part of future work.

4. Data Sources, Curation, and Preprocessing

4.1. Chapter Overview

This chapter describes how heterogeneous chemical, toxicological, and pharmacokinetic data were assembled into a modeling-ready resource for the Neural PK-PD framework. The emphasis is not only on data acquisition, but also on the scientific role of each source, the distinction between real and synthetic components, and the harmonization steps required before feature engineering and model training.

The sources used in this thesis fall into three categories. First, *real training data* are obtained from TDC and ChEMBL and provide the endpoint labels used in the baseline harmonized multi-task model. Second, *auxiliary mechanistic data* are obtained from PK-DB and support the pharmacokinetic interpretation of downstream ODE simulations. Third, *illustrative reference data* are drawn from PubChem and a ToxCast/Tox21-style synthetic reference set, which are used to contextualize safety screening and data-collection strategy rather than to define the final supervised learning targets.

4.2. Data Sources and Scientific Role

4.2.1. PubChem: Assays and Exemplar Compounds

PubChem (NCBI; https://pubchem.ncbi.nlm.nih.gov/) was used as an open-access reference source for public bioassay and compound-structure data. Its role in this thesis is illustrative rather than central to the final multi-task training set: it provides representative screening assays and example molecular structures that help situate the project within an early-discovery workflow.

Two assay families were retained for reference purposes: an hERG qHTS screen relevant to cardiac safety, and a CYP3A4 inhibition assay relevant to metabolic interaction risk. In addition, 3D SDF structures for warfarin, midazolam, and caffeine were downloaded as exemplar compounds for structural inspection and method prototyping. PubChem therefore supports the thesis primarily as a source of screening context and chemically interpretable examples, rather than as the main source of model-development labels.

4.2.2. PK-DB: Pharmacokinetic Studies and Parameters

PK-DB (https://www.pk-db.com/; REST API v1) was selected because it provides curated pharmacokinetic studies with hierarchical study structure, summary outputs, and concentration-time observations. Unlike cross-sectional ADMET datasets, PK-DB captures temporal pharmacokinetic behavior and therefore provides the mechanistic bridge between predicted molecular properties and compartmental PK-PD simulation.

The extracted records include study-level metadata, population structure, intervention information, parameter summaries, and concentration-time measurements. Parameters such as clearance (CL), volume of distribution (Vd), area under the curve (AUC), peak concentration (Cmax), half-life, absorption rate, and bioavailability are particularly important because they inform the interpretation and calibration of the downstream ODE framework. In this thesis, PK-DB is used as an auxiliary mechanistic resource rather than as a direct source of supervised labels for the multi-task neural predictor.

4.2.3. TDC: Standardized ADMET Benchmarks

Therapeutics Data Commons (TDC; https://tdcommons.ai/) provides standardized molecular machine-learning benchmarks with canonical splits and well-documented endpoint definitions. TDC forms the core real-data backbone for the classification and regression tasks in the final predictive model because it offers consistent benchmark structure, literature grounding, and direct comparability with prior work.

The TDC resources used here span absorption, distribution, metabolism, and toxicity endpoints. Across the eight downloaded benchmark files, the current raw-source total is 20,995 rows. Among these, the hERG inhibition, Caco-2 permeability, and hepatocyte clearance datasets are the most important for the baseline harmonized framework because they map directly onto safety, absorption, and elimination behavior in the PK-PD pipeline. Other TDC datasets are retained to provide broader ADMET context during exploratory analysis and dataset characterization.

4.2.4. ChEMBL: Binding Affinity Data

ChEMBL (EBI; https://www.ebi.ac.uk/chembl/) is the primary source of binding-affinity information in the thesis and provides the molecular activity data needed for the pharmacodynamic component of the framework. Its importance lies in the fact that binding potency is the endpoint most directly linked to downstream effect modeling through the EC_{50} term in the PD formulation.

Two ChEMBL-derived resources are distinguished explicitly. The first is a *real* curated set of canonical SMILES and measured activity values collected for eight pharmacologically relevant targets under consistent activity-type constraints (IC50, Ki, or EC50 with pChEMBL annotation). The second is a *synthetic* binding-affinity reference set constructed to illustrate target coverage and potency-distribution behavior during early exploratory analysis. The synthetic set is useful for controlled visualization and prototype experiments, but it should be interpreted as supportive rather than definitive evidence.

For the real ChEMBL subset, only measured activities were retained, canonical SMILES were validated, duplicate compounds were merged, and descriptors such as molecular weight and LogP were recomputed from structure rather than copied uncritically from source annotations. In the current workspace, the downloaded ChEMBL files comprise 2,000 rows in the exploratory synthetic binding set and 3,410 rows in the real SMILES binding set.

4.2.5. ToxCast/Tox21-Style Toxicity Reference Data

ToxCast and Tox21 are important public programs for high-throughput toxicity screening, but in the present work they are represented through a synthetic reference set designed to reflect the structure of broad toxicity-screening data rather than to serve as a final supervised learning target. The purpose of including this material is to illustrate how multi-category safety information can be organized and interpreted in an integrated preclinical pipeline.

The reference set covers cardiac, developmental, nuclear receptor, hepatic, renal, stress-response, and metabolic toxicity categories with tiered severity labels. This provides a useful conceptual counterpart to the focused hERG toxicity endpoint used in the final model: whereas hERG gives a specific cardiotoxicity label, the ToxCast/Tox21-style synthetic reference demonstrates the broader safety-screening landscape into which such predictions would ultimately be deployed.

Toxicity Category	Hit Rate	Risk Level
Cardiac toxicity (e.g., QT prolongation)	25%	HIGH
Developmental/Reproductive toxicity	15%	CRITICAL
Nuclear receptor activation	30%	HIGH
Liver toxicity	22%	HIGH
Kidney toxicity	12%	MEDIUM
Stress response (e.g., ROS, HSP)	20%	HIGH
Metabolic effects (glucose/lipid)	18%	MEDIUM

Table 4.1: Representative toxicity categories used to illustrate multi-domain safety screening in a ToxCast/Tox21-style reference dataset.

4.3. Data Curation and Harmonization

Because the selected sources differ substantially in structure, scale, and scientific purpose, raw download alone is insufficient for model development. A harmonization stage is therefore required to transform heterogeneous source files into a consistent analytical resource.

The curation logic used throughout the chapter follows four principles. First, structural identifiers are standardized by canonicalizing SMILES strings and excluding chemically invalid entries. Second, endpoint definitions are harmonized so that binding data, toxicity labels, permeability outcomes, and clearance values can be interpreted consistently across sources. Third, duplicate compounds are resolved before model splitting to avoid optimistic bias. Fourth, molecular descriptors are recomputed from structure where necessary so that downstream features are derived uniformly rather than inherited from mixed source conventions.

This distinction between *source data* and *analysis-ready data* is important for thesis clarity. The scientific contribution lies not in downloading files from public repositories, but in converting them into a coherent multi-task resource with explicit task definitions, traceable provenance, and defensible quality control.

4.4. Summary of Source Roles in the Framework

Table 4.2 summarizes the role of each source in the thesis. Rather than combining incompatible record counts into a single total, the table emphasizes the scientific contribution of each source and whether it contributes real training labels, auxiliary mechanistic context, or illustrative reference material.

No.	Source	Primary Role	Status in Thesis
1	PubChem	Screening context and exemplar structures	Illustrative reference data
2	PK-DB	Pharmacokinetic studies and time-course interpretation	Auxiliary mechanistic data
3	TDC ADMET	hERG, Caco-2, clearance, and benchmark context	Real training data
4	ChEMBL	Binding affinity and target-specific activity data	Real training data + synthetic reference set
5	ToxCast/Tox21-style set	Multi-category toxicity-screening illustration	Synthetic reference data

Table 4.2: Scientific role of each data source in the thesis framework.

4.5. Data Validation and Quality Assurance

4.5.1. Validation Checks in the Pipeline

The download pipeline includes automatic validation at each stage:

1. **API response validation**: JSON parsing with fallback for structurally inconsistent responses (e.g., PK-DB API variations).
2. **File integrity**: Downloaded files logged with size in KB for rapid verification.
3. **SMILES canonicalization**: RDKit validation ensures chemical validity. Invalid SMILES (e.g., unbalanced parentheses, unknown atom symbols) are rejected.
4. **Molecular descriptor reliability**: MW and logP computed via RDKit; graceful degradation if RDKit unavailable (values set to NaN with warning).
5. **Duplicate handling**: Per-source and global deduplication applied (e.g., CHeMBL global SMILES deduplication with activity averaging).
6. **Record counts**: Summary statistics logged for each source for manual inspection.

4.5.2. Critical Files for Downstream Modeling

The pipeline verifies that the following files exist before proceeding to feature engineering:

1. `tdc_herg.csv` – hERG toxicity labels (classification).

2. `tdc_caco2_wang.csv` – Caco-2 permeability labels (classification).

3. `tdc_clearance_hepatocyte_az.csv` – Metabolic clearance labels (regression).

4. `chembl_binding_affinity.csv` – Binding affinity, synthetic (regression).

5. `chembl_binding_affinity_with_smiles.csv` – Binding affinity, real SMILES (regression).

4.6. Exploratory Data Analysis and Data Quality Assessment

4.6.1. Dataset Loading and Composition

Following source acquisition and basic validation, an exploratory data analysis (EDA) stage was conducted to assess whether the downloaded resources were sufficiently coherent for downstream modeling. The purpose of this stage was not to define the final harmonized training matrix directly, but to inspect coverage, distributional behavior, endpoint plausibility, and potential quality issues before harmonization. For that reason, the counts shown in this section should be interpreted as source-level or benchmark-level summaries, whereas the baseline harmonized task counts used for model training are defined later in the multi-task assembly subsection.

Source	**Dataset Type**	**Records**	**Primary Use**
PubChem	Bioassay screening (hERG, CYP3A4)	5,000	Toxicity filtering
ChEMBL	Binding affinity data	2,000	Pharmacodynamic potency
TDC ADMET	Three benchmark subsets used in EDA	2,778	ADMET endpoint inspection
ToxCast-style set	Aggregated priority-tier reference rows	570,167	Safety-screening illustration
PK-DB	Pharmacokinetic studies + time-courses	796 studies	ODE interpretation

Table 4.3: Source-level overview used during exploratory analysis. Counts refer to the specific benchmark subsets or aggregated reference files inspected in this section and are distinct from the later harmonized task counts used for model training.

4.6.2. ChEMBL Binding Affinity Analysis

The ChEMBL exploratory subset provides the pharmacodynamic activity landscape for the thesis. At the EDA stage, the aim is to verify that the retrieved activity values span a chemically meaningful potency range and cover more than one target family, rather than to optimize the exact training subset.

The observed activity range extends from weak to highly potent compounds, giving the model access to a broad potency spectrum instead of a narrow local structure-activity regime. This is important because downstream pharmacodynamic simulation depends on converting predicted potency into an EC_{50}-like quantity; a narrow or highly truncated activity range would make that mapping much less informative. In addition, the eight selected targets span kinases, GPCRs, and nuclear receptors, which gives the binding task some mechanistic diversity rather than confining it to a single assay family.

Figure 4.1 serves an illustrative purpose here. The left panel shows the overall potency distribution, while the right panel shows that target coverage is not concentrated entirely in one protein class. Taken together, these diagnostics support the use of ChEMBL as the primary source of pharmacodynamic supervision in the later multi-task model.

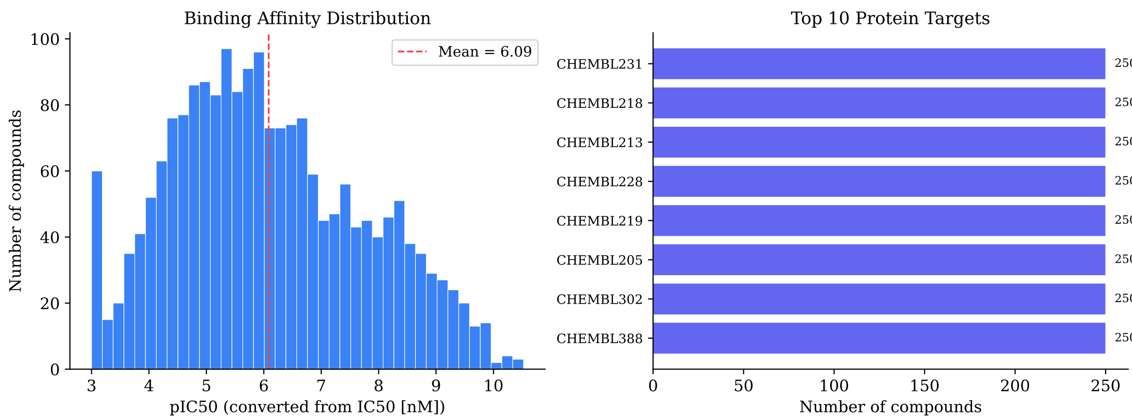

Figure 4.1: ChEMBL Binding Affinity Distribution and Target Coverage. Left panel: histogram of pIC50 values (converted from IC50 [nM]) across 2,000 compounds, mean pIC50 = 6.09, with mean overlay (red dashed line). Right panel: top 10 pharmacological targets by compound frequency (kinases, GPCRs, nuclear receptors).

4.6.3. ToxCast Toxicity Assessment

The ToxCast-style reference set is used in this chapter as a broad safety-screening illustration rather than as a direct supervised learning target. Its value lies in showing how toxicological evidence can be stratified by severity and mechanism, thereby complementing the much narrower but operationally central hERG endpoint used later in the predictive model.

Risk Level Distribution:

Risk Level	Count	Percentage	Interpretation
CRITICAL	18,927	5.7%	Severe toxicity hits requiring exclusion
HIGH	142,654	42.9%	Significant hazard; dose escalation caution
MEDIUM	109,863	33.1%	Moderate risk; manageable with strategy
LOW	61,063	18.4%	Acceptable safety profile
Total	**332,507**	**100%**	–

Table 4.4: ToxCast Risk Level Distribution across 332,507 Assay Results.

The primary categories represented are cardiac, developmental, nuclear receptor, hepatic, renal, stress-response, and metabolic toxicity. This breadth is useful because it reminds the reader that real preclinical safety assessment is multi-domain even when the final machine-learning framework focuses on a smaller set of tractable endpoints. In this subsection, two complementary views are used: the table above summarizes one representative toxicity reference file (332,507 rows), whereas Figure 4.2 aggregates the critical-, high-, and representative-priority files into a broader descriptive visualization (570,167 rows). The figure therefore functions as a contextual safety illustration rather than as the definition of a single modeling dataset.

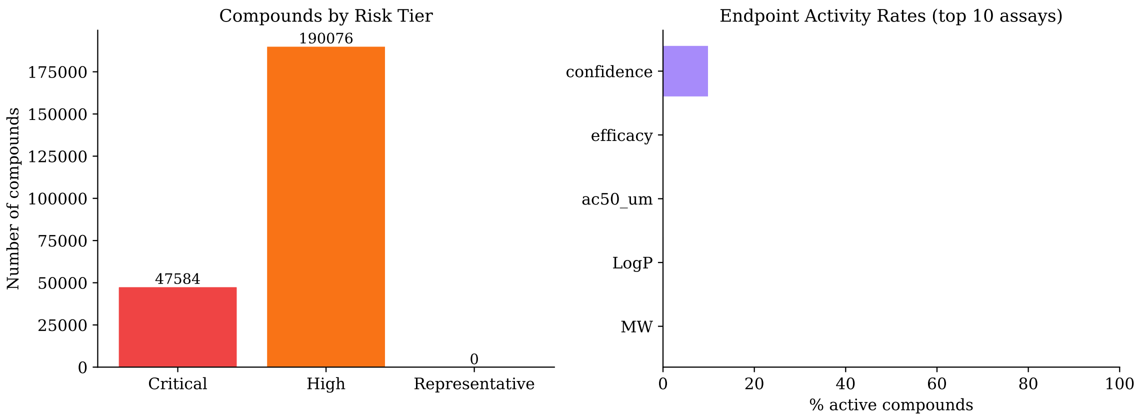

Figure 4.2: Illustrative toxicity risk stratification in the aggregated ToxCast-style reference set (570,167 rows across critical-, high-, and representative-priority files). Left panel: compound counts across priority tiers. Right panel: activity rates across the most prevalent toxicity assay endpoints. The figure summarizes broad safety-screening structure rather than defining a final supervised training target.

4.6.4. TDC ADMET Properties Analysis

Three TDC benchmarks are examined in detail because they provide the endpoints that map most directly into the PK-PD framework: hERG inhibition for safety, Caco-2 permeability for absorption, and hepatocyte clearance for elimination. At this stage, the goal is to inspect benchmark-level behavior and confirm that each endpoint exhibits a plausible distribution for later modeling.

hERG Inhibition (Cardiotoxicity Risk)

The hERG endpoint captures a clinically important liability because potassium-channel blockade is strongly associated with QT prolongation and pro-arrhythmic risk. The benchmark subset used for exploratory visualization contains 655 compounds and remains sufficiently balanced to support meaningful binary discrimination, making AUROC an appropriate primary metric later in the thesis. The figure panel is intended to show class balance at the benchmark level; the final harmonized task counts used in model training are reported separately in the dataset-assembly subsection.

Caco-2 Permeability (Intestinal Absorption)

The Caco-2 benchmark is retained because it provides an experimentally grounded proxy for intestinal absorption. Its distribution is heterogeneous rather than degenerate, which is desirable for classification after thresholding. In pharmacological terms, high permeability suggests good oral absorption potential, while low permeability flags compounds that may require formulation support or alternative administration routes. The middle panel of Figure 4.3 illustrates this spread at the raw-benchmark level.

Hepatocyte Clearance (Metabolic Stability)

The clearance benchmark provides the elimination-related endpoint required for the PK component of the framework. The benchmark subset used for exploratory visualization contains 1,213 compounds. Its right-skewed distribution is pharmacologically plausible: many compounds exhibit moderate stability, while a smaller subset is cleared rapidly enough to imply reduced exposure duration. This endpoint is therefore especially valuable because it links a learned molecular property directly to the elimination term in the downstream ODE model.

Figure 4.3 should be read as an exploratory benchmark summary rather than a declaration of final model-training counts. The plotted sample sizes reflect the benchmark subsets used for visualization, while the integrated task counts used for supervised learning are reported later after harmonization and deduplication.

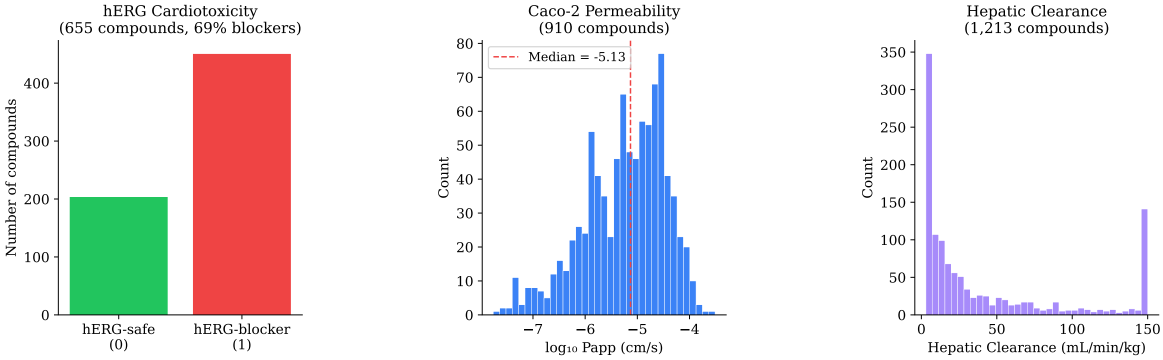

Figure 4.3: Exploratory benchmark distributions for the three TDC endpoints most relevant to the PK-PD framework. Left: hERG class balance in the benchmark subset used for visualization. Middle: Caco-2 permeability distribution. Right: hepatocyte-clearance distribution showing heterogeneous metabolic stability. These panels illustrate endpoint behavior prior to final harmonized dataset assembly.

The importance of these three TDC endpoints is that they map directly onto safety, absorption, and elimination constraints in the PK-PD simulation module. In other words, they are not merely generic ADMET benchmarks; they are the endpoints through which the multi-task predictor becomes mechanistically relevant.

4.6.5. Pharmacokinetic Studies (PK-DB)

PK-DB plays a different role from ChEMBL and TDC. It is not used primarily to define a high-volume supervised prediction task, but to anchor the mechanistic interpretation of the framework. Its hierarchical records, pharmacokinetic summaries, and concentration-time observations provide the temporal context that cross-sectional ADMET datasets lack. This makes PK-DB essential for showing how molecular predictions can be translated into compartmental PK behavior rather than remaining isolated scalar outputs.

4.6.6. Data Quality and Integrity Assessment

The exploratory analyses above were followed by explicit data-quality checks to determine whether the retrieved sources were suitable for harmonization and downstream learning.

1. **Missingness**: critical modeling columns in the ChEMBL and TDC sources exhibit low

missingness, making descriptor recomputation and target assignment feasible without aggressive imputation.

2. **Outlier behavior**: the observed tails in potency, permeability, and clearance are consistent with plausible pharmaceutical variation rather than obvious transcription error. Where needed, extreme values are flagged and handled conservatively rather than removed indiscriminately.
3. **Descriptor plausibility**: recomputed molecular descriptors fall predominantly within drug-like ranges, indicating that canonicalization and descriptor extraction are functioning as intended.
4. **Label consistency**: endpoint encodings are sufficiently stable to support harmonization into common regression and classification tasks.
5. **Bias awareness**: no obvious evidence of catastrophic source distortion was observed at the exploratory stage, although this does not eliminate the broader issue of assay heterogeneity across public data sources.

4.6.7. Multi-Task Learning Dataset Assembly

The objective of harmonization is realized in the baseline multi-task dataset assembly used in the exploratory analysis and feature-engineering workflow. Because the contributing sources differ in scale, endpoint definition, and intended scientific role, the final training matrix is not a simple concatenation of raw files. Instead, it is constructed by aligning a shared molecular representation with task-specific labels.

1. **Shared molecular representation**: molecular weight (MW), octanol-water partition coefficient (LogP), and 2,048-bit Morgan fingerprints define the common feature space. In the current executed notebook, real 2,048-bit Morgan fingerprints are available for the TDC-derived tasks, while the ChEMBL binding subset remains a zero-fingerprint fallback because canonical SMILES are not present in the current exploratory snapshot.
2. **Task-specific labels**:
   (a) Binding affinity (regression): continuous pIC50 values from ChEMBL.
   (b) hERG inhibition (binary classification): TDC hERG labels (inhibitor = 1, non-inhibitor = 0).
   (c) Caco-2 permeability (binary classification): median-binarized TDC Caco-2 permeability.

(d) Hepatocyte clearance (regression): continuous TDC clearance values.

3. **Baseline harmonized task counts**:

   (a) Total rows in the harmonized matrix: 4,768.

   (b) Binding affinity: 2,000 samples.

   (c) hERG inhibition: 655 samples.

   (d) Caco-2 permeability: 900 samples.

   (e) Hepatocyte clearance: 1,213 samples.

   (f) Feature dimensionality: 2,050 (2 physicochemical descriptors + 2,048 fingerprint bits).

This assembly stage marks the transition from exploratory source inspection to model-ready supervision. It is at this point, rather than at the raw-source stage, that the thesis obtains the definitive baseline task counts used in the exploratory analysis and feature-engineering workflow. Later experimental chapters describe subsequent data upgrades and expanded structural inputs applied in later notebooks.

4.6.8. Feature Preprocessing and Normalization

Before neural-network training, the harmonized feature matrix is preprocessed to ensure numerical stability and cross-task comparability.

1. **Descriptor completeness**: records missing MW or LogP are excluded to maintain complete low-dimensional physicochemical inputs.

2. **Scaling**: MW and LogP are standardized to zero mean and unit variance. Binary fingerprint bits remain unscaled so that structural presence/absence information is preserved directly.

3. **Verification**: post-scaling checks confirm that the standardized descriptors are numerically well behaved and suitable for joint optimization with the fingerprint inputs.

4. **Target handling**:

   (a) Regression targets (binding affinity and clearance) are standardized for optimization but remain interpretable in their original scientific units after inverse transformation.

   (b) Classification targets (hERG and Caco-2) remain binary and are handled directly by the classification loss.

5. **Persisted output**: the processed feature matrix is stored as `phase2_multitask_features_with_fingerprints.csv` so that all subsequent experiments operate on the same frozen inputs.

4.7. Feature Engineering

4.7.1. Molecular Representations and Feature Space Design

Feature engineering in this thesis is designed to support both predictive accuracy and mechanistic interpretability. The chosen representation must be expressive enough for molecular property prediction, while remaining sufficiently structured to support shared learning across four heterogeneous endpoints.

Raw SMILES strings are therefore transformed into complementary feature spaces:

1. **Extended-connectivity fingerprints (ECFP/Morgan fingerprints)**: 2,048-bit binary vectors computed with radius 2 capture local structural motifs such as atom environments, connectivity patterns, and common functional groups. This fingerprint representation provides a compact but information-rich summary of molecular structure and serves as the primary shared input to the baseline multi-task network.

2. **Physicochemical Descriptors**:

   (a) Molecular weight (MW): Typically in pharmaceutical range 150–550 Da.

   (b) Octanol-water partition coefficient (logP): Lipophilicity measure (typically -2 to +5).

   (c) Topological polar surface area (TPSA): Polarity and permeability predictor.

   (d) Hydrogen bond donors (HBD) and acceptors (HBA): Counts relevant for absorption and solubility (Lipinski's rule of five).

   (e) Rotatable bond count (RotBonds): Molecular flexibility metric.

   (f) Aromatic ring count: Relevant for metabolic stability.

   These descriptors are computed from canonical SMILES using RDKit and complement the sparse fingerprint signal with interpretable global molecular properties.

3. **Graph representation**: for graph-based experiments, molecules are additionally encoded as atom-bond graphs so that message-passing architectures can learn relational structure directly rather than relying only on fixed fingerprints.

4. **Binding-context features**: for ChEMBL-specific experiments, target identity can be included explicitly so that the model can distinguish among target classes and binding regimes.

The key design decision is to combine a strong generic representation (Morgan fingerprints) with a small set of interpretable physicochemical descriptors. This gives the baseline model structural specificity without making it dependent on heavy task-specific featurization.

4.7.2. Task-Specific Feature Sets and Multi-Task Integration

The multi-task framework uses a shared feature encoder followed by task-specific output heads. This design is appropriate because the four endpoints are biologically distinct but chemically related: the same molecular substructures can influence potency, permeability, safety, and clearance in different ways.

Prediction Task	**Features**	**Samples**	**Target Type**
Binding affinity (Reg.)	Morgan FP + MW + LogP	2,000	Continuous pIC50
hERG inhibition (Cls.)	Morgan FP + MW + LogP	655	Binary (0/1)
Caco-2 permeability (Cls.)	Morgan FP + MW + LogP	900	Binary (0/1)
Hepatocyte clearance (Reg.)	Morgan FP + MW + LogP	1,213	Continuous (mL/min/kg)

Table 4.5: Baseline multi-task learning setup for the harmonized feature-engineering matrix. Later chapters report upgraded dataset variants.

The shared encoder produces a latent representation that is reused across tasks, while the task-specific heads specialize that representation for either regression or classification. This separation between shared and task-specific learning is central to the thesis because it supports positive transfer without forcing all endpoints into the same output geometry.

Task-specific heads then branch from the shared bottleneck:

1. **Regression heads** (binding, clearance): Linear output layer with continuous predictions; optimized via mean-squared error (MSE) loss.

2. **Classification heads** (hERG, Caco-2): Sigmoid output layer with logistic loss; optimized via binary cross-entropy.

This architecture is intended to capture the practical reality of molecular discovery data: structure is shared, but supervision is partial and endpoint-specific.

4.7.3. Docking-Derived Bridge Signal (RapidDock Prototype)

In selected fusion experiments, an optional bridge signal derived from a RapidDock-inspired prototype is appended to the baseline molecular representation. This signal is intended to capture coarse geometric information related to protein-ligand compatibility and is used only as an exploratory augmentation, not as a mandatory component of the core thesis model.

4.7.4. Feature Storage and Reusability

The complete preprocessed feature matrix is persisted to disk as `phase2_multitask_features_with_fingerprints.csv`. This is methodologically important because it freezes the exact inputs used in later phases and makes ablation, retraining, and reproducibility checks possible without regenerating descriptors from raw source files.

Persisting the matrix supports:

1. **Reproducibility**: Exact same features used across all experimental runs and model variants.
2. **Efficient loading**: Model training pipelines load pre-computed features directly without re-computing RDKit descriptors or fingerprints.
3. **Ablation studies**: Features can be subsetted (e.g., exclude fingerprints, include only descriptors) for diagnostic comparisons.
4. **External validation**: Features can be exported and used in external tools or by collaborators.

4.7.5. Hyperparameter Configuration

The feature engineering and model training pipeline are controlled through a centralized configuration, ensuring that all experimental runs use a consistent and reproducible set of parameters. The selected configuration fixes the molecular representation, network capacity, optimization schedule, task-loss weighting, and data split proportions used throughout the feature-engineering notebook. The complete specification is summarized in Table 4.6.

The input representation is defined by a 2,048-bit Morgan fingerprint combined with two physicochemical descriptors (molecular weight and LogP), yielding a total input dimensionality

Parameter	Value	Description
`n_bits`	2048	Morgan fingerprint bits (radius=2)
`physico_dim`	2	MW and LogP physicochemical descriptors
`input_dim`	2050	Total feature dimensionality
`hidden_dim`	128	Shared encoder hidden layer size
`latent_dim`	64	Bottleneck representation dimension
`dropout`	0.2	Encoder dropout rate
`reg_head_hidden`	64	Regression head hidden width
`reg_head_dropout`	0.1	Regression head dropout
`batch_size`	64	Mini-batch size
`epochs`	300	Maximum training epochs
`learning_rate`	10^{-3}	Adam optimizer learning rate
`weight_decay`	10^{-4}	L2 regularization
`patience`	40	Early stopping patience
`grad_clip`	1.0	Gradient clipping norm
`w_binding`	1.8	Binding affinity loss weight
`w_hERG`	1.0	hERG inhibition loss weight
`w_caco2`	2.0	Caco-2 permeability loss weight
`w_clearance`	1.0	Clearance loss weight
`w_physics`	0.1	Physics-informed ODE loss weight
`focal_gamma`	2.0	Focal loss gamma (classification)
`herg_pos_weight`	2.5	hERG positive class weight
`caco2_pos_weight`	1.0	Caco-2 positive class weight
`test_size`	0.20	Test set fraction
`val_size`	0.10	Validation set fraction

Table 4.6: Feature-engineering notebook hyperparameter configuration for the baseline harmonized matrix. Binding and Caco-2 receive elevated loss weights to stabilize optimization of the harder and less homogeneous tasks in multi-task training.

of 2,050. The shared encoder uses a hidden width of 128 and a latent bottleneck of 64, which provides sufficient capacity to learn cross-task molecular structure relationships while maintaining a compact representation suitable for regularized multi-task training.

The optimization parameters were chosen to balance convergence speed and training stability. In particular, the learning rate of 10^{-3}, weight decay of 10^{-4}, early-stopping patience of 40 epochs, and gradient clipping at 1.0 were used to reduce overfitting and prevent unstable updates during joint optimization of heterogeneous endpoints.

Task-specific loss weights were introduced to prevent the shared encoder from drifting toward whichever endpoint was easiest to optimize. Binding affinity and Caco-2 permeability therefore receive elevated weights of 1.8 and 2.0, respectively, reflecting their greater heterogeneity and more limited effective signal in the baseline harmonized matrix. For the classification endpoints, focal loss with $\gamma = 2.0$ down-weights easy examples and concentrates learning on difficult decision boundaries, while the positive-class weight of 2.5 for hERG addresses the greater cost of missing potentially cardiotoxic compounds.

Finally, the physics-informed term ($w_{\text{physics}} = 0.1$) acts as a soft ODE-consistency regularizer. Its role is not to replace supervised learning, but to bias the model toward predictions that remain compatible with the downstream pharmacokinetic formulation. In this way, the selected hyperparameter configuration supports both predictive performance and mechanistic coherence across the integrated PK-PD workflow.

4.7.6. Data Quality Gate

Before model fitting, the harmonized feature matrices are subjected to a concrete validation gate implemented in the feature-engineering workflow. The purpose of this gate is to ensure that no downstream result is inflated by preventable preprocessing artifacts.

1. **NaN/Inf detection**: Any feature or target column with missing or infinite values is flagged and removed. Records with $> 5\%$ missing values across feature columns are excluded.

2. **Duplicate feature rows**: Exact-duplicate molecular fingerprints (verified via MD5 hash of feature vectors) are removed, keeping only the first occurrence. Deduplication statistics are logged per task.

3. **Split leakage audit**: After train/validation/test splitting, exact feature-row overlap between splits is computed. Any non-zero overlap count terminates pipeline execution and requires investigation before training proceeds.

4. **Class balance diagnostics**: For binary classification tasks (hERG, Caco-2), positive class fractions in each split are compared. Imbalance exceeding a factor of 2.0 between splits triggers a resampling or re-stratification step.

5. **Target distribution stability**: Regression task mean and standard deviation are compared between splits using a Kolmogorov-Smirnov test ($p > 0.05$ threshold). Significant distribution shifts indicate non-representative splits.

This quality gate converts general data-quality principles into executable acceptance criteria. It therefore serves as the bridge between descriptive curation and defensible model evaluation.

4.8. Quality Control and Split Strategy

4.8.1. Quality Control Procedures

To ensure that downstream model performance reflects genuine predictive signal rather than artifacts of preprocessing or partitioning, a structured quality-control procedure is applied before any training run. These checks address numerical validity, duplicate handling, split integrity, class balance, and descriptor plausibility.

1. **Missing-value and numerical-validity checks**: all task-specific matrices are screened for NaN and infinite values. Records failing these checks are excluded, imputed where justified, or masked on a task-specific basis.

2. **Duplicate resolution**: exact duplicates and canonical-SMILES duplicates are identified prior to splitting. For binding affinity data, repeated measurements are merged by activity averaging; for classification tasks, consensus labels are assigned.

3. **Leakage auditing**: train, validation, and test partitions are examined to confirm that no molecule appears in more than one split. Scaffold-based diagnostics are additionally used to assess whether structurally similar compounds are disproportionately shared across partitions.

4. **Class-balance diagnostics**: for hERG and Caco-2, label prevalence is reported explicitly so that AUROC, F1, and threshold-dependent metrics can be interpreted in the correct imbalance context.

5. **Descriptor plausibility checks**: molecular properties such as molecular weight, LogP, and topological polar surface area are compared against plausible pharmaceutical ranges.

Outliers are flagged for inspection but are not removed automatically unless there is evidence of annotation or preprocessing error.

Taken together, these procedures reduce the risk that reported model performance is driven by data artifacts rather than by learnable molecular signal.

4.8.2. Train/Validation/Test Split Strategy

The integrated dataset is partitioned into training, validation, and test subsets using a randomized but controlled split strategy. The objective is to preserve label balance where required, maintain strict separation between subsets, and ensure that reported results are reproducible across runs.

1. **Stratification**: for classification endpoints, the hERG and Caco-2 label distributions are preserved across splits to avoid distorted prevalence in validation and test evaluation.
2. **Leakage prevention**: splitting is performed only after SMILES-level deduplication, and post-split overlap audits verify that no feature row or canonical molecule is shared across partitions. Scaffold-based diagnostics remain available as an additional sensitivity check for structural similarity.
3. **Split proportions**: the default feature-engineering notebook configuration uses 70% training, 10% validation, and 20% test, consistent with the centralized hyperparameter settings described in Section 5.6.5. TDC benchmark datasets retain their official predefined partitions where such splits are provided.
4. **Random seed control**: NumPy and PyTorch seeds are fixed so that split generation, training, and metric estimation remain reproducible.

This split design provides a practical balance between statistical efficiency and evaluation rigor. The training set remains large enough for stable representation learning, while the validation and test sets remain strictly isolated for model selection and final reporting.

4.8.3. Metadata and Provenance Tracking

Each dataset is accompanied by a JSON metadata record documenting its origin, structure, and known limitations. The stored metadata include:

1. Source (API, version, download date).
2. Record count and summary statistics.

3. License and citation information.
4. Column descriptions and data types.
5. Known limitations or data quality notes.

This metadata layer supports reproducibility, provenance tracking, and compliance with source-specific licensing and citation requirements. It also preserves an audit trail linking each processed dataset back to its acquisition context, transformation steps, and modeling assumptions.

5. Methodology and Model Architecture

5.1. Exploratory Data Analysis Methods

5.1.1. Descriptive Statistical Analysis

For each data source, comprehensive descriptive statistics were computed:

1. **Continuous variables**: Mean, median, standard deviation, min/max, and quartile values calculated using numpy and scipy. Distributions assessed visually via histograms and Q-Q plots.

2. **Categorical variables**: Value counts and frequency distributions computed for all nominal features (target class, toxicity risk level, target protein name). Chi-square tests applied to assess label balance in classification datasets.

3. **Missing data assessment**: For each dataset, missing value patterns documented by column. Records with $> 50\%$ missing values excluded; remaining missing values handled via task-specific strategies (deletion for critical columns, imputation for ancillary features).

4. **Outlier detection**: Univariate outliers identified via z-score ($|z| > 3$) and interquartile-range (IQR $\times 1.5$) methods. Multivariate outliers detected via Mahalanobis distance on principal components. Outliers flagged but retention decision made based on domain knowledge (e.g., high-clearance compounds retained as valid, not artifacts).

5.1.2. Distribution Visualization

Matplotlib-based visualizations generated for all endpoints:

1. **Univariate distributions**: Histograms with kernel density estimation overlays to assess modality, skewness, and kurtosis. Overlay of mean (red dashed line) and median (optional) to diagnose skew.

2. **Categorical distributions**: Bar charts (vertical for counts, horizontal for rankings) with explicit value labels. Color-coding applied to encode secondary information (e.g., risk level severity in ToxCast).

3. **Correlation structure**: Pairwise scatter plots between molecular descriptors (MW vs. logP) to assess feature independence and detect potential multicollinearity.

4. **Comparison across sources**: Side-by-side plots for same endpoint measured in different datasets (e.g., TDC hERG vs. PubChem hERG) to assess consistency.

5.1.3. Data Quality Checks

Systematic verification applied to all datasets:

1. **SMILES validation**: RDKit-based validation of all SMILES strings. Invalid SMILES (unbalanced parentheses, unrecognized atoms, valence errors) flagged and excluded.

2. **Molecular descriptor plausibility**: Computed MW, logP checked against pharmaceutical known ranges. Implausible values (e.g., MW < 50 Da or > 1000 Da, logP < -5 or > 10) flagged for manual review.

3. **Label consistency**: For datasets with multiple assay types, activity values standardized to common scale (pIC50 for binding, binary 0/1 for classification). Conflicting labels identified and resolved via majority voting or expert curation.

4. **Duplicate detection**: Exact-match and SMILES-canonicalization-based duplicate identification. For binding data, duplicate activities merged via averaging. For classification, consensus labels computed.

5. **Statistical tests for bias**: Kolmogorov-Smirnov (K-S) tests applied to compare feature distributions between train/validation/test splits. No significant differences ($p > 0.05$) expected in valid random stratified splits.

5.2. Multi-Task Learning Architecture

5.2.1. Model Design Notebook Data Interface

The implementation in `phase3b_model_design.ipynb` does not begin from raw tensors alone, but from the persisted feature-engineering artifacts. These artifacts provide task-specific arrays, task types, feature metadata, and scaling objects, after which the notebook augments the feature space with a RapidDock-derived `docking_quality` bridge signal for the binding task. The binding matrix is rebuilt from the real ChEMBL SMILES file and extended to 2,051 input features, while the hERG, Caco-2, and clearance tasks receive a padded `docking_quality`=0.0 column so that all tasks share a common schema.

After augmentation, the notebook reconstructs per-task train, validation, and test DataLoaders. Feature matrices are standardized separately for each task using `StandardScaler`; regression targets are also standardized, whereas classification targets remain binary. This design keeps the training pipeline consistent across heterogeneous endpoints while preserving the interpretation of logits for classification tasks.

5.2.2. Shared Encoder Design

The model-design notebook baseline uses a single shared encoder followed by task-specific heads. In the implemented notebook, the encoder receives a 2,051-dimensional input vector consisting of molecular weight, logP, 2,048 Morgan fingerprint bits, and the appended `docking_quality` feature. The encoder is defined as:

Block 1	`Linear(input_dim, hidden_dim)` → `LayerNorm(hidden_dim)` → `ReLU` → `Dropout(config['dropout'])`
Block 2	`Linear(hidden_dim, hidden_dim)` → `LayerNorm(hidden_dim)` → `ReLU` → `Dropout(config['dropout'])`
Latent projection	`Linear(hidden_dim, latent_dim)` → `ReLU`

With the current model-design notebook configuration, this corresponds to an encoder of size 2,051 → 128 → 128 → 64. Layer normalization is used instead of batch normalization because training proceeds through interleaved task batches of different sizes; under this regime, LayerNorm is more stable than BatchNorm, whose running statistics can become unreliable.

5.2.3. Task-Specific Heads

Four task-specific heads branch from the 64-dimensional latent representation:

1. **Binding affinity head** (`DeepRegressionHead`): a deeper multilayer perceptron, 64 → 64 → 32 → 1, with ReLU activations and dropout on the hidden path. This additional depth is used because binding prediction is empirically the most demanding regression task.

2. **hERG head** (`ClassificationHead`): a compact classifier, 64 → 32 → 1, returning a raw logit rather than a post-sigmoid probability.

3. **Caco-2 head** (`ClassificationHead`): architecturally identical to the hERG head and likewise producing logits.

4. **Clearance head** (`RegressionHead`): a compact regression head, $64 \rightarrow 32 \rightarrow 1$, used for the simpler scalar clearance task.

The use of raw logits for the classification heads is important because the training criterion applies `BCEWithLogits`-based losses directly, while the sigmoid transformation is deferred to evaluation and threshold selection.

5.2.4. Multi-Task Objective

At the conceptual level, the architecture optimizes a weighted sum of task-specific objectives:

$$\mathcal{L}_{\text{task}} = w_{\text{bind}}\mathcal{L}_{\text{binding}} + w_{\text{hERG}}\mathcal{L}_{\text{hERG}} + w_{\text{Caco-2}}\mathcal{L}_{\text{Caco-2}} + w_{\text{clear}}\mathcal{L}_{\text{clearance}} \tag{5.1}$$

In the notebook implementation, this base task loss is instantiated as weighted MSE for regression tasks and a logits-based classification loss for hERG and Caco-2, with focal-loss modulation enabled by default. The full model-design notebook loss additionally includes a physics penalty on the predicted PK trajectories, described in Section 5.3.

5.3. Full Multi-Task Neural ODE Architecture

The model-design notebook (`phase3b_model_design.ipynb`) implements the complete `MultiTaskPKPDMo` establishing the baseline architecture for all subsequent experiments. It loads feature-engineering artifacts and adds a RapidDock bridge feature before training.

5.3.1. Objectives and Target Metrics

Task	Metric	Target
Binding affinity (regression)	R^2	> 0.60
hERG inhibition (classification)	AUROC	> 0.80
Caco-2 permeability (classification)	AUROC	> 0.75
Hepatic clearance (regression)	RMSE	< 1.0

Table 5.1: Model-design notebook target metrics. The architecture is evaluated against these thresholds to determine readiness for the fine-tuning notebook.

5.3.2. RapidDock Bridge Feature Augmentation

Before model construction, the model-design notebook augments the feature matrix with a docking quality signal derived from the RapidDock prototype (`structure_binding_pkpd_bridge.csv`).

A per-target mean docking score is computed for the 8 ChEMBL protein targets and appended as feature 2051:

- **Binding task**: each compound's feature vector gains `docking_quality` = the mean RapidDock distance-biased transformer score for its ChEMBL target.
- **ADMET tasks** (hERG, Caco-2, clearance): `docking_quality` is padded to 0.0 since these assays lack a single protein-target geometry context equivalent to the 8-target bridge.
- **Result**: input dimensionality updated from 2,050 to **2,051** and `config[`input_dim']` correspondingly updated.

5.3.3. Layer-by-Layer Architecture

SharedEncoder (`input_dim` → `hidden_dim` → `hidden_dim` → `latent_dim`):

- Block 1: `Linear(2051, 128) -- LayerNorm(128) -- ReLU -- Dropout(config['dropout'])`
- Block 2: `Linear(128, 128) -- LayerNorm(128) -- ReLU -- Dropout(config['dropout'])`
- Bottleneck: `Linear(128, 64) -- ReLU`

LayerNorm is used in preference to BatchNorm because interleaved multi-task batches of different sizes destabilize BatchNorm's running statistics.

Task-specific heads branching from the 64-dimensional bottleneck:

- **DeepRegressionHead** (binding): a $64 \to 64 \to 32 \to 1$ multilayer perceptron with ReLU activations and 0.1 dropout.
- **ClassificationHead** (hERG, Caco-2): a $64 \to 32 \to 1$ head producing raw logits; sigmoid is applied only at inference and loss evaluation.
- **RegressionHead** (clearance): a $64 \to 32 \to 1$ regression head.

PKODEFunc (Neural ODE pharmacokinetics):

- **PK parameter network**: a $64 \to 32 \to 2$ network with Tanh nonlinearity predicts $[\log \widehat{CL}, \log \widehat{V}]$; outputs are exponentiated to ensure positivity.
- **ODE dynamics**: $\frac{dC}{dt} = -\frac{\widehat{CL}}{\widehat{V}} \cdot C$ (first-order elimination)
- **Inference**: the prediction routine calls `torchdiffeq.odeint` over a user-specified time grid.

The complete `MultiTaskPKPDModel` forward pass routes each batch to the appropriate head via a `task` argument; passing `task='all'` returns all four task predictions and the latent representation simultaneously.

5.3.4. Training Loop Design

Training uses **interleaved task sampling at minimum-batch epoch sizing**:

1. Each epoch processes $n_{\text{steps}} = \min_k |\mathcal{L}_k|$ steps, where $|\mathcal{L}_k|$ is the number of batches for task k. This prevents the smallest task (Caco-2, ≈ 10 batches) from cycling repeatedly within a single epoch.

2. At each step, one gradient update is performed per task in round-robin order: `binding` $\rightarrow$ `hERG` $\rightarrow$ `Caco-2` $\rightarrow$ `clearance`.

3. DataLoader `shuffle=True` ensures different samples are drawn each epoch, providing effective data augmentation without explicit augmentation code.

Optimizer: Adam with lr $= 10^{-3}$, weight_decay $= 10^{-4}$, gradient clipping at norm $= 1.0$. **Scheduler**: ReduceLROnPlateau with factor $= 0.5$ and patience $= 8$ (monitors total validation loss). **Early stopping**: patience $= 40$ epochs on total validation loss; best weights saved to `best_model.pt`.

5.3.5. Loss Function

The multi-task loss extends Equation 5.1 with a physics-informed penalty:

$$\mathcal{L}_{\text{total}} = w_{\text{bind}}\, \mathcal{L}_{\text{MSE}}^{\text{bind}} + w_{\text{hERG}}\, \mathcal{L}_{\text{focal}}^{\text{hERG}} + w_{\text{Caco-2}}\, \mathcal{L}_{\text{focal}}^{\text{Caco-2}} + w_{\text{clear}}\, \mathcal{L}_{\text{MSE}}^{\text{clear}} + w_{\text{phys}}\, \mathcal{P} \tag{5.2}$$

The physics penalty $\mathcal{P}$ discourages biologically implausible PK curves:

$$\mathcal{P} = \underbrace{\mathbb{E}[\max(0,\, -C(t))]}_{\text{negative concentration}} + \underbrace{\mathbb{E}[\max(0,\, C(t+1) - C(t))]}_{\text{concentration increase}} \tag{5.3}$$

The focal classification loss for class-imbalanced tasks uses per-sample down-weighting of easy examples:

$$\mathcal{L}_{\text{focal}} = \sum_i w_i \cdot (1 - p_t^{(i)})^{\gamma} \cdot \text{BCE}(z_i, y_i), \quad \gamma = 2.0 \tag{5.4}$$

where w_i is the sample-dependent class weight induced by the configured positive-class weight, and $p_t^{(i)} = \exp(-\text{BCE}(z_i, y_i))$ is the sample-level confidence estimate. In the current notebook, hERG uses an elevated positive-class weight and Caco-2 uses its own configured class weight, with both heads trained on raw logits rather than post-sigmoid probabilities.

5.3.6. Validation and Threshold Selection

Validation uses Youden's J-statistic to select per-task decision thresholds on the validation split:

$$\theta^* = \arg\max_{\theta}\big[\mathrm{TPR}(\theta) - \mathrm{FPR}(\theta)\big]$$

This threshold is reported alongside AUROC and accuracy at each validation epoch, and is used for per-epoch F1 reporting. The final production threshold is re-calibrated in the fine-tuning notebook.

Model-design artifacts (model weights, scalers, raw arrays, config) are saved to `data/processed/phase3b_artifacts/` for loading by the fine-tuning notebook.

5.4. Fine-Tuning, Calibration, and Leakage Mitigation

The fine-tuning notebook (`phase3c_finetuning.ipynb`) loads the persisted feature-engineering and model-design artifacts, reconstructs the baseline model and task-specific data loaders, and then applies task-specific fine-tuning, threshold calibration with guardrails, production threshold selection, and split-leakage mitigation. In the implemented notebook, these artifact and checkpoint paths are resolved from the nearest ancestor containing the shared `data/` directory, so the workflow remains executable even if the notebooks are reorganized within the `Coding/` workspace.

5.4.1. Task-Specific Fine-Tuning (Variants 1 and 2)

Variant 1: Heads-Only Fine-Tuning

All parameters except the two classification heads are frozen:

- **Frozen**: SharedEncoder, PKODEFunc, DeepRegressionHead (binding), RegressionHead (clearance)
- **Trainable**: `herg_head` and `caco2_head` only ($\ll 1\%$ of total parameters)

Configuration: lr $= 3 \times 10^{-4}$, weight_decay $= 10^{-5}$, 40 epochs, patience $= 10$. The scheduler monitors mean validation AUROC (mode='max').

Variant 2: Partial Encoder Unfreeze

A controlled variant that additionally unfreezes the final encoder block:

- **Frozen**: PKODEFunc, regression heads, encoder blocks 0–7
- **Trainable**: Encoder blocks 8–9 (final `Linear(128,64)` + `ReLU`), `herg_head`, `caco2_head`

Configuration: lr $= 10^{-4}$ (lower to protect partially-frozen representations), 60 epochs, patience = 12. The best of Variant 1 and Variant 2 by validation mean AUROC is selected as the calibration base model.

5.4.2. Threshold Calibration

Unconstrained Threshold Sweep

For each classification task, a grid of 91 thresholds $\theta \in [0.05, 0.95]$ is evaluated on the validation set; the threshold maximizing F1 is locked:

$$\hat{\theta} = \arg\max_{\theta \in \Theta} \mathrm{F1}\big(y_{\mathrm{val}},\ \mathbf{1}[p_{\mathrm{val}} \geq \theta]\big)$$

Constrained Threshold Calibration

To avoid degenerate low-threshold policies (which inflate F1 while collapsing operational behavior), a guardrail-constrained sweep enforces minimum accuracy and precision:

- hERG guardrails: accuracy ≥ 0.50, precision ≥ 0.12
- Caco-2 guardrails: accuracy ≥ 0.50, precision ≥ 0.50

The constrained threshold $\hat{\theta}_c$ is the F1-maximizing threshold that satisfies both guardrails on the validation set.

Production Threshold Policy

The production threshold is selected per task via an automatic decision rule:

1. If the constrained threshold $\hat{\theta}_c$ yields higher test F1 *and* maintains accuracy $\geq \theta^{\mathrm{acc}}_{\mathrm{min}}$: use $\hat{\theta}_c$.
2. Otherwise: revert to baseline $\theta = 0.5$.

This policy ensures operational robustness: no classification threshold is locked that reduces prediction reliability below acceptable accuracy bounds. Production thresholds are saved to `data/processed/phase3c_artifacts/production_thresholds.json`.

5.4.3. Split-Leakage Mitigation

The feature-engineering quality gate (Section 4.7.6) detected feature-row overlap between train, validation, and test splits, primarily in the binding task. The fine-tuning notebook remediates this:

1. **MD5-based deduplication**: For each task's feature matrix, an MD5 hash is computed per row; only the first occurrence of each hash is retained. Duplicate rows that would be trivially learnable (memorized across splits) are removed.

2. **Quality gate re-validation**: After deduplication, the full 5-step quality gate is re-run on the cleaned splits to confirm zero leakage across all tasks.

3. **Retraining**: The complete model is retrained from random initialization on clean deduplicated splits with recalculated class weights (`config_clean`).

4. **Comparison against March 8 baseline**: Final locked-threshold metrics on clean splits are compared against a recorded March 8 reference (hERG AUROC = 0.4836, Caco-2 AUROC = 0.4719, binding R^2 = 0.0019, clearance RMSE = 0.9766) to quantify the combined benefit of real fingerprints, fine-tuning, and leakage remediation.

Fine-tuning artifacts (production model weights, thresholds, clean loader state, feature metadata, raw arrays, and final report metrics) are saved to `data/processed/phase3c_artifacts/` for loading by the experiment notebook.

5.5. Fingerprint Upgrade: From Synthetic to Real SMILES

5.5.1. Motivation for Fingerprint Upgrade

Initial experiments used a lightweight 256-bit Morgan representation that provided limited structural resolution, and the exploratory ChEMBL binding snapshot remained structurally uninformative because canonical SMILES were not available in that file. The representation-upgrade step therefore increased fingerprint resolution to 2,048 bits and rebuilt the harmonized matrix from the real TDC benchmark SMILES while preserving the ChEMBL binding subset as a zero-fingerprint fallback. This upgrade is reported as the structural-resolution transition stage before later thesis-essential real-binding integration experiments.

In this current notebook-aligned representation-upgrade stage, the 2,048-bit Morgan fingerprints directly encode structural information for the TDC tasks, including:

- Functional groups (hydroxyl, carbonyl, amine)
- Ring systems (aromatic, aliphatic)
- Bond connectivity patterns and heteroatom environments

5.5.2. 2048-Bit Real-SMILES Upgrade

Property	Baseline Encoding	Upgraded Encoding
Fingerprint bits	256	2048
SMILES type	Lightweight / partially uninformative baseline	Real TDC SMILES + ChEMBL zero-fallba
Input dimensionality	258	**2050**
Binding samples	2,000	2,000
hERG samples	655	655
Caco-2 samples	910	900
Clearance samples	1,213	1,213

Table 5.2: Representation-upgrade comparison aligned with the current executed notebook. The upgrade increases fingerprint resolution to 2,048 bits and introduces real structural signal for the TDC-derived tasks, while the exploratory ChEMBL binding subset remains a zero-fingerprint fallback because SMILES are not available in the current snapshot.

5.5.3. Current Notebook-Aligned Representation Upgrade

The current Phase 2 notebook does not yet replace the exploratory ChEMBL binding subset with a real-SMILES binding file. Instead, it performs the following notebook-aligned upgrade:

- ChEMBL binding subset: 2,000 compounds with MW, LogP, and pChEMBL targets; fingerprint columns remain zero because SMILES are not present in the snapshot used here.
- TDC hERG subset: 655 compounds with real SMILES and 100% non-zero 2,048-bit Morgan fingerprints.
- TDC Caco-2 subset: 900 compounds after outlier filtering and median binarization, with 100% non-zero 2,048-bit Morgan fingerprints and a 50% positive rate.
- TDC clearance subset: 1,213 compounds with real SMILES and 100% non-zero 2,048-bit Morgan fingerprints.

- Persisted output: a 4,768-row processed matrix stored as `phase2_multitask_features_with_fingerprints.csv`, containing 2,050 model input features plus task and target metadata columns.

5.6. Fine-Tuning Strategy

Two fine-tuning variants are evaluated: heads-only updates and selective encoder unfreezing. Threshold calibration is applied post-training for classification heads.

5.6.1. hERG AUROC Push (hidden_dim=256 + pos_weight sweep)

To close the gap toward the 0.80 AUROC target for hERG cardiotoxicity prediction, the following strategy was implemented:

1. **Wider model**: hidden_dim increased from 128 to 256, providing additional representational capacity for the 4,746 hERG training samples.
2. **pos_weight sweep**: Grid search over $\{0.5, 1.0, 1.5, 2.0, 3.0\}$ to identify the optimal positive class weight that maximizes validation hERG AUROC.
3. **Extended patience**: Increased from 40 to 60 epochs to allow the wider network sufficient time to converge.

The sweep used 60 epochs with patience=15 for efficiency, followed by full training (300 epochs, patience=60) with the winning configuration.

5.7. Graph Encoder and Fusion Architecture

5.7.1. Pure-PyTorch GNN Molecular Encoder

A message-passing neural network (MPNN) replaces 2048-bit Morgan fingerprints for the binding affinity prediction branch, built directly on PyTorch and RDKit atom/bond features without external GNN libraries.

Atom features extracted via RDKit:

- Atomic number (one-hot encoded over $\{C, N, O, S, P, F, Cl, Br, I, \text{other}\}$)
- Degree (number of bonded neighbors)

- Formal charge
- Number of hydrogens
- Aromaticity flag

Bond features:

- Bond type: single, double, triple, aromatic
- Ring membership flag

Architecture: 3-layer GCN with dense adjacency representation and masking. Each message-passing step aggregates atom neighborhood information via:

$$h_v^{(k)} = \mathrm{ReLU}\left(W^{(k)} \cdot \mathrm{AGG}\left(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\} \cup \{h_v^{(k-1)}\}\right)\right) \tag{5.5}$$

where $\mathcal{N}(v)$ is the neighborhood of atom v and AGG is mean pooling. The final graph representation is obtained via global mean pooling over all atom embeddings.

5.7.2. End-to-End Joint GNN and MLP Fusion

The fusion architecture concatenates the GNN graph embedding with the MLP fingerprint latent representation before the final task head:

$$z_{\text{fusion}} = \mathrm{concat}\left(z_{\text{GNN}},\ z_{\text{MLP}}\right) \in \mathbb{R}^{d_{\text{GNN}}+d_{\text{MLP}}} \tag{5.6}$$

This allows the model to leverage both learnable structural graph information (GNN branch) and dense fingerprint-derived features (MLP branch), with the fusion head learning to weight their relative contributions. Both branches are trained jointly with a shared binding regression loss.

5.8. Neural ODE PK-PD Integration

5.8.1. Mechanistic Simulation Design

Predicted molecular endpoints are mapped to PK-PD parameters and used in ODE simulation to derive concentration-time and effect-time profiles. The mapping from predictions to ODE parameters is:

Two-compartment PK ODE system (IV bolus administration):

$$\frac{dC_c}{dt} = -(k_e + k_{12})\, C_c + k_{21}\, C_t + u(t) \tag{5.7}$$

Predicted Property	PK/PD Role	Parameter
Hepatic clearance (regression)	Elimination rate constant	$k_e = \mathrm{CL}/V_c$
Caco-2 permeability (probability)	Oral absorption rate	k_a
Binding affinity pChEMBL (regression)	Pharmacodynamic potency	EC_{50}
hERG probability (classification)	Cardiac safety flag	Safety threshold

Table 5.3: Mapping of predicted molecular properties to ODE pharmacokinetic/pharmacodynamic parameters.

$$\frac{dC_t}{dt} = k_{12}\, C_c - k_{21}\, C_t \tag{5.8}$$

where C_c and C_t are the central and peripheral compartment concentrations, k_{12} and k_{21} are inter-compartmental transfer rate constants, k_e is the elimination rate constant derived from predicted clearance, and $u(t)$ is the dosing function.

Pharmacodynamic Emax model (effect-site linkage):

$$E(t) = E_{\max} \cdot \frac{C_c(t)}{EC_{50} + C_c(t)} \tag{5.9}$$

where $E_{\max}$ is the maximal pharmacological effect, and EC_{50} is derived from the predicted binding affinity (pChEMBL $\to EC_{50}$ via $EC_{50} = 10^{-\mathrm{pChEMBL}}$).

The ODEs are numerically integrated using `torchdiffeq.odeint` (Dormand-Prince RK45 solver), enabling gradient-based optimization through the ODE solution if required for physics-informed training.

5.9. Probability Calibration

5.9.1. Temperature Scaling and Platt Scaling

Classification models are post-hoc calibrated to ensure that predicted probabilities faithfully reflect empirical frequencies. Two calibration methods are evaluated:

1. **Temperature Scaling**: Divides logit scores by a scalar temperature $T > 0$ before sigmoid transformation:

$$\hat{p} = \sigma(z/T), \quad T > 0 \tag{5.10}$$

   The temperature T is optimized on the validation set via negative log-likelihood (NLL) minimization. $T > 1$ softens overconfident predictions; $T < 1$ sharpens underconfident outputs.

2. **Platt Scaling**: Applies an affine transformation to the logit:

$$\hat{p} = \sigma(a \cdot z + b) \tag{5.11}$$

Parameters (a, b) are learned by logistic regression on validation set logits and labels.

Calibration evaluation: Expected Calibration Error (ECE) is used as the primary metric:

$$\text{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left| \text{acc}(B_m) - \text{conf}(B_m) \right| \tag{5.12}$$

where B_m are calibration bins (10 bins of equal width), and $\text{acc}(B_m)$, $\text{conf}(B_m)$ are the accuracy and average predicted confidence within bin m. Lower ECE indicates better calibration.

5.10. RapidDock-Inspired Alternative Branch

As an alternative method track, a RapidDock-inspired distance-biased transformer is developed as a prototype branch (`rapid_dock_paper_aligned_prototype.ipynb`). Unlike the main MLP/GNN pipelines which predict scalar ADMET endpoints, this branch models the *geometry* of a protein-ligand complex directly by predicting pairwise atom distances and then reconstructing 3D ligand coordinates.

5.10.1. Input Representation and Data Pipeline

The prototype operates on joint protein-ligand representations rather than molecular fingerprints:

- **Ligand representation**: Real ChEMBL SMILES from the binding-affinity file are converted to 3D conformers via RDKit `EmbedMultipleConfs` (3 conformers, MMFF optimization). Mean coordinates over conformers are used to reduce conformer-dependent noise. Atom types are encoded as 8-class one-hot vectors based on atomic number modulo 8.

- **Protein representation**: Synthetic context points (48–96 atoms) sampled near the ligand centroid with Gaussian noise ($\sigma = 8.0$ Å). Features are random 32-dimensional vectors. This is a prototype approximation; real implementations use protein residue embeddings from PDB structures.

- **Input filtering**: Only ligands with 8–20 heavy atoms and successful RDKit 3D embedding are retained, yielding approximately 80–120 valid complexes from 140 sampled SMILES.

- **Train/validation split**: 83% train, 17% validation (random split, reproducible via `np.random.default_rng(42)`).

Distance targets:

- $\mathbf{D}_{\text{LL}} \in \mathbb{R}^{L\times L}$: pairwise Euclidean distances between all pairs of ligand atoms.
- $\mathbf{D}_{\text{LP}} \in \mathbb{R}^{L\times P}$: pairwise distances from each ligand atom to each protein context point.
- $\mathbf{D}_{\text{joint}} \in \mathbb{R}^{(L+P)\times(L+P)}$: full joint distance matrix used as attention bias.

5.10.2. Distance-Biased Multi-Head Attention

The core architectural innovation is the **distance-biased MHA** layer. Standard scaled dot-product attention is modified to incorporate pairwise physical distances as negative additive bias before the softmax:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{head}}}} - \alpha \cdot \mathbf{D}_{\text{joint}}\right) V \tag{5.13}$$

where $\alpha = 0.25$ is the bias scale and $\mathbf{D}_{\text{joint}}$ is the joint protein-ligand distance matrix (broadcast over heads). This implements the intuition that spatially distant atom pairs should receive lower attention weight, encoding prior geometric knowledge directly into the attention mechanism.

Padded tokens (variable-length batching) are masked to $-\infty$ before softmax, and the output is zeroed for padded positions.

5.10.3. Architecture: RapidDockPrototype

The full model (`RapidDockPrototype`) consists of:

1. **Projection layers**: Separate linear projections for ligand features ($8 \to 128$) and protein features ($32 \to 128$).
2. **Type embeddings**: Learnable $\mathbb{R}^{128}$ vectors for token type (0 = ligand, 1 = protein), added to projected features before the transformer.
3. **4 × DBTransformerBlock**: Each block contains:
   - LayerNorm + Distance-Biased MHA (8 heads, $d_{\text{head}} = 16$)
   - LayerNorm + feed-forward MLP block
   - Residual connections around both sub-layers; padded positions zeroed after each block.

4. **Final LayerNorm** applied to all tokens.

5. **Distance prediction heads**: $\widehat{\mathbf{D}}_{\mathrm{LL}} = \texttt{cdist}(h_L, h_L)$ and $\widehat{\mathbf{D}}_{\mathrm{LP}} = \texttt{cdist}(h_L, h_P)$, where h_L, h_P are the ligand and protein output token representations. Distance outputs are computed via Euclidean distance in the learned representation space, with no explicit normalization.

5.10.4. Symmetry-Aware Loss

A key challenge in ligand distance prediction is that chemically equivalent atoms (same element, similar bonding environment) can be permuted without changing the molecular identity. Naively assigning fixed atom indices causes the loss to penalize physically equivalent solutions.

The symmetry-aware loss enumerates permutations of repeat-type atom groups and takes the minimum-loss permutation:

$$\mathcal{L}_{\mathrm{sym}} = \min_{\pi\in\Pi}\left[\frac{\sum_{i,j} |\hat{D}^{\mathrm{LL}}_{ij} - D^{\mathrm{LL}}_{\pi(i)\pi(j)}| \cdot \mathbf{1}[D^{\mathrm{LL}}_{\pi(i)\pi(j)} < \delta]}{\sum_{i,j} \mathbf{1}[D^{\mathrm{LL}} < \delta]} + \frac{\sum_{i,k} |\hat{D}^{\mathrm{LP}}_{ik} - D^{\mathrm{LP}}_{\pi(i)k}| \cdot \mathbf{1}[D^{\mathrm{LP}} < \delta]}{\sum_{i,k} \mathbf{1}[D^{\mathrm{LP}} < \delta]}\right] \tag{5.14}$$

where Π is the set of permutations over same-type atom groups, $\delta = 20\,\text{Å}$ is a distance cutoff (only short-to-medium range contacts contribute to the loss), and the minimum is over up to 24 enumerated permutations. Long-range pairs ($> 20\,\text{Å}$) are excluded to focus learning on chemically relevant contact geometry.

5.10.5. Ligand Coordinate Reconstruction (L-BFGS)

Given predicted distance matrices $\hat{\mathbf{D}}_{\mathrm{LL}}$ and $\hat{\mathbf{D}}_{\mathrm{LP}}$ from the trained model and the known protein context coordinates, ligand 3D coordinates are recovered by L-BFGS minimization:

$$\hat{\mathbf{X}}_L = \arg\min_{\mathbf{X}\in\mathbb{R}^{L\times 3}}\left[\sum_{i<j} \big|\|\mathbf{x}_i - \mathbf{x}_j\| - \hat{D}^{\mathrm{LL}}_{ij}\big| + \sum_{i,k} \big|\|\mathbf{x}_i - \mathbf{p}_k\| - \hat{D}^{\mathrm{LP}}_{ik}\big|\right] \tag{5.15}$$

(masked to $\delta = 20\,\text{Å}$ contacts). Optimization is initialized near the weighted protein centroid (weights from softmax of mean predicted LP distances) and run for 200 L-BFGS steps with strong Wolfe line search. The reconstruction quality is measured by RMSD between reconstructed and ground-truth ligand coordinates (no alignment applied, since the prototype uses a single canonical orientation per compound).

For reproducibility, the prototype uses a fixed random-seed train/validation split (83%/17%) consistent with the main pipeline's split governance, and is trained under the same implementation framework (PyTorch-based optimization and fixed random seeds). The primary optimization target is binding geometry reconstruction, with validation loss monitored across training epochs.

This branch is intentionally scoped as exploratory: hyperparameter search is limited, calibration and uncertainty procedures are not fully integrated, and the branch is therefore reported as a feasibility pathway rather than a production candidate that replaces the main MLP and GNN+MLP tracks.

6. Experimental Setup and Protocol

6.1. Environment and Reproducibility

All experiments were implemented in Python 3.14 within a controlled scientific-computing environment. The core software stack was selected to cover molecular preprocessing, neural network training, numerical simulation, and statistical evaluation in a single reproducible workflow. The principal package versions used in the project are listed below:

- **PyTorch** 2.10.0: Neural network training, GPU acceleration (MPS/CUDA), ODE integration
- **RDKit** 2025.9.5: Molecular fingerprint computation, SMILES canonicalization, atom/bond feature extraction
- **torchdiffeq** 0.2.5: Differentiable ODE solver (Dormand-Prince RK45)
- **scikit-learn** 1.8.0: Preprocessing, metrics, calibration
- **pandas** 3.0.0, **numpy** 2.4.1: Data management and numerical operations

Reproducibility was enforced through fixed random seeds (`SEED` = 42) applied consistently to NumPy, PyTorch, and Python's `random` module. The complete package environment is recorded in `requirements.txt`, allowing the experimental stack to be recreated without relying on implicit local configuration.

Beyond package pinning and seed control, reproducibility in this thesis depends on an explicit staged artifact interface. The feature-engineering notebook exports harmonized feature matrices and preprocessing metadata; the model-design notebook exports the baseline model state together with loader metadata; the fine-tuning notebook exports the production model and calibrated decision thresholds; and the experiment notebook consumes these persisted artifacts for simulation, optimization, and decision analysis. This notebook-to-notebook contract reduces hidden state between notebooks and ensures that later analyses operate on versioned outputs rather than on regenerated intermediate data.

The notebook workflow further improves robustness by resolving the nearest shared `data/` root dynamically, which makes the project less sensitive to notebook relocation within the `Coding/` directory while preserving a stable `data/processed/` and `data/outputs/` structure. In addition, lightweight runtime execution logs record step start and completion times during long-running experiments. These logs do not alter the computational results, but they provide an audit trail that improves traceability for the fine-tuning and experiment notebooks.

6.2. Training Protocol

All models are trained using the Adam optimizer with:

- Learning rate: 10^{-3}
- Weight decay (L2): 10^{-4}
- Gradient clipping: max norm = 1.0 (prevents exploding gradients)
- Early stopping: patience = 40 epochs (extended to 60 for the hERG improvement sweep)
- Learning rate scheduling: ReduceLROnPlateau with factor 0.5 and patience 10
- Batch size: 64
- Maximum epochs: 300

Classification tasks use focal loss ($\gamma = 2.0$) to handle class imbalance. Regression tasks use mean squared error (MSE). The combined multi-task loss is a weighted sum (Equation 6.1) with task-specific weights from Table 4.6.

6.3. Experimental Sequence

The thesis-essential experiment notebook program was organized as a staged progression from data realism, through predictive refinement, to mechanistic and translational evaluation. Rather than presenting every exploratory notebook branch with equal emphasis, the main thesis sequence focuses on the experiments that most directly strengthened predictive performance, improved decision reliability, and enabled downstream PK-PD analysis. This yields a clearer narrative: first, the molecular representation and binding data were upgraded; second, the predictive models were refined and calibrated; third, the resulting predictions were integrated into a mechanistic PK-PD framework; and finally, the framework was used for multi-objective regimen selection and virtual clinical evaluation.

In practice, the core sequence begins with real binding-data integration and task-specific hERG improvement, because these steps establish the strongest predictive foundation for later analysis. Probability calibration is then applied to ensure that classification outputs can be interpreted as reliable decision variables rather than only rank-based scores. After that, a joint GNN + MLP fusion model is evaluated as the strongest structure-aware predictor for binding affinity. These predictive outputs are then propagated into the Neural ODE PK-PD layer, which supports efficacy-safety trade-off analysis, uncertainty quantification, and trial-level simulation. Exploratory branches such as standalone GNN testing, one-factor scenario sweeps, and single-objective optimization are retained as supporting analyses, but are not treated as the main thesis path because they are either intermediate ablations or are superseded by more defensible multi-objective analyses.

The representation-upgrade experiment, the standalone GNN ablation, and the scenario, sensitivity, and single-objective optimization analyses remain useful for internal comparison, but they are better treated as supporting or exploratory analyses. The representation-upgrade stage mainly serves as a transitional step before full real-data integration. The standalone GNN is informative as an architectural ablation, yet it is less important than the final fusion model. The scenario, sensitivity, and single-objective optimization analyses help illustrate system behavior, but the single-objective regimen search is ultimately less defensible than the Pareto-front formulation. For this reason, the main thesis argument is built around real binding integration, hERG-focused refinement, probability calibration, joint fusion modeling, Neural ODE PK-PD integration, Pareto analysis, Monte Carlo validation, and virtual clinical trial simulation.

Experiment	Thesis-essential experiment and role
Experiment 1	Later real ChEMBL binding SMILES and full 2048-bit fingerprint integration; establishes the transition from the notebook-aligned TDC-focused upgrade to a realistic binding-input representation.
Experiment 2	hERG AUROC improvement via hidden_dim expansion and positive-class weighting; defines the strongest toxicity-classification configuration carried forward into later analysis.
Experiment 3	Probability calibration using temperature scaling and Platt scaling with ECE evaluation; converts raw classifier outputs into decision-relevant risk probabilities.
Experiment 4	End-to-end joint GNN + MLP fusion model; provides the strongest structure-aware binding model and the final predictive architecture for downstream translation.
Experiment 5	Neural ODE PK/PD integration with population simulation; links predictive outputs to mechanistic exposure-response behavior.
Experiment 6	Pareto-front efficacy-safety analysis; replaces single-objective optimization with a clinically interpretable multi-objective trade-off framework.
Experiment 7	Monte Carlo and bootstrap uncertainty analysis; quantifies robustness of the recommended regimens and validates stability of the simulated conclusions.
Experiment 8	In-silico clinical trial simulation and operating characteristics; demonstrates how the final framework can support translational dose-selection decisions.

Table 6.1: Revised Phase 3D thesis-essential experimental sequence. The main narrative emphasizes the experiments that most directly improved predictive quality, calibration, mechanistic interpretability, and translational decision support. Intermediate exploratory analyses are discussed only as supporting evidence.

6.4. RapidDock Prototype Protocol

The RapidDock-inspired branch (`rapid_dock_prototype.ipynb`) is executed as a constrained prototype study following a defined experimental protocol:

1. **Dataset**: 140 SMILES sampled from the ChEMBL binding affinity dataset (deterministic subset via `np.random.default_rng(42)`). After RDKit 3D embedding and atom-count filtering, approximately 80–120 valid protein-ligand complexes are retained.
2. **Optimizer**: Adam, $\text{lr} = 2 \times 10^{-4}$, default PyTorch betas (no weight decay).
3. **Training duration**: 30 epochs, no early stopping (prototype runtime is short enough to train fully).
4. **Batch size**: 8 complexes per batch with variable-length padding and masking.
5. **Loss**: Symmetry-aware distance L1 loss (Equation 5.10.4), averaged over valid (unpadded) ligand-protein pairs.
6. **Evaluation**: Validation loss reported every 5 epochs; final reconstruction RMSD evaluated on the first validation example without alignment correction.
7. **Hyperparameter search**: Not performed for this prototype; all architectural and training hyperparameters are fixed at their initial values.
8. **Reproducibility**: `torch.manual_seed(42)` and `np.random.seed(42)` applied at notebook start.

7. Results

7.1. Chapter Overview

This chapter presents the empirical results as a focused progression from baseline multi-task prediction toward translational PK-PD decision support. The main narrative follows the thesis-essential sequence defined in Chapter 6: a transitional notebook-aligned representation upgrade, later real-data integration, target-specific predictive improvement, probability calibration, structure-aware fusion modeling, mechanistic PK-PD simulation, multi-objective trade-off analysis, uncertainty quantification, and virtual clinical evaluation. Earlier or intermediate branches are retained only where they establish a necessary comparison point or explain why a later modeling decision was adopted.

Accordingly, the baseline and notebook-aligned fingerprint-upgrade results are treated primarily as reference stages rather than as the final contribution of the study. The central emphasis is placed on the later sequence formed by real binding-data integration, hERG-focused refinement, calibration of the safety classifiers, the final joint GNN+MLP architecture, Neural ODE PK-PD integration, Pareto-based efficacy-safety analysis, Monte Carlo robustness checks, and in-silico clinical trial simulation. This organization ensures that the chapter reflects the final thesis argument rather than the full exploratory notebook chronology.

7.2. Baseline Multi-Task Predictive Performance

The Phase 3B model establishes the starting point for all later comparisons. It combines a shared encoder, four task-specific heads, and a differentiable PK component into a unified `MultiTaskPKPDModel`. Training was stable and the architecture successfully handled mixed regression and classification targets within a common optimization loop. The baseline is therefore technically valid as a common reference model, even though its predictive accuracy is limited.

The main weakness of the baseline is that it operates before the full molecular representation upgrade. In this configuration, binding discrimination remains near zero and hERG classification is close to random discrimination, indicating that the model has insufficient structural signal to learn robust structure-property relationships. These baseline results are therefore not presented as competitive final performance, but as the reference point against which the value of later representation, architecture, and simulation improvements can be quantified.

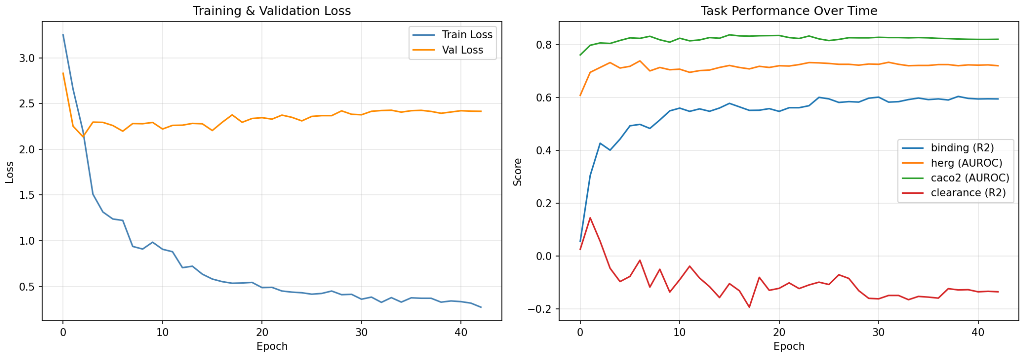

Figure 7.1: Training and validation loss curves for the baseline multi-task model. The smooth decrease in both curves confirms stable optimization of the shared encoder and task-specific heads, including the differentiable PK component.

7.3. Transitional Molecular Representation Upgrade

The first major improvement examined in this study was the transition from a low-resolution baseline representation to a 2,048-bit fingerprint encoding derived from real molecular structures where SMILES were available. In the current Phase 2 notebook, this structural upgrade is realized for the TDC-derived hERG, Caco-2, and clearance tasks, while the exploratory ChEMBL binding subset remains descriptor-only with zero-fingerprint fallback. This experiment functions mainly as a transitional reference stage before the later thesis-essential real-binding and calibration sequence, but it remains important because it isolates the effect of richer structural encoding even before substantial architectural changes are introduced.

This finding indicates that the early performance ceiling was driven at least in part by limited molecular representation rather than by insufficient model complexity. The fingerprint upgrade transforms the problem from one dominated by only a few physicochemical descriptors into one in which structure-activity patterns can be learned directly. As a result, the representation upgrade becomes the new reference point for all subsequent experiments.

Property	Baseline Encoding	Upgraded Encoding
Fingerprint bits	256	2048
SMILES source	Limited / partially uninformative baseline	Real TDC structures + ChEMBL zero-fallback
Input dimensionality	258	2050
Primary role	Minimal structural signal	Stronger structural SAR signal for TDC tasks

Table 7.1: Summary of the molecular representation upgrade aligned with the current notebook state. The 2,048-bit encoding introduces real structural signal for the TDC-derived tasks while the exploratory ChEMBL subset remains a zero-fingerprint fallback.

7.4. Real Binding Data Integration

Although the fingerprint upgrade improves the structural informativeness of the harmonized matrix, binding affinity remains the endpoint most sensitive to data realism because the current exploratory ChEMBL subset still lacks SMILES in this notebook stage. The next experiment therefore remains the explicit real-binding integration step, where descriptor-only ChEMBL placeholders are replaced by a structurally faithful binding dataset. This separation is methodologically important because it distinguishes the TDC-focused representation upgrade from the later binding-data realism intervention.

Scientifically, this result matters because it shows that the model benefits not only from richer descriptors, but also from more faithful endpoint-aligned information. The gains in binding prediction are larger than the corresponding gains in the other tasks, which is expected: the added information is most directly relevant to affinity modeling. This confirms that the framework is sensitive to the realism of the underlying molecular evidence and that improvements in one task need not transfer uniformly to all other tasks.

7.5. hERG Refinement and Final Fusion Model

Once the input representation and binding data were improved, the next question was whether structure-aware learning could further enhance performance. A graph neural network encoder was introduced to learn directly from molecular topology, followed by a joint GNN+MLP fusion model that combines graph-derived embeddings with fingerprint-based latent features. The graph-based model outperformed the fingerprint-only baseline on binding affinity, and the fused model delivered the strongest final predictive performance.

These results indicate that graph and fingerprint representations are complementary rather than redundant. The GNN branch captures relational and local structural context, while the MLP branch retains dense tabular information from the molecular fingerprint space. Their integration produces a stronger predictive model than either branch alone. For this reason, the joint GNN+MLP fusion architecture is treated as the final predictive model of the thesis, while the GNN-only model is retained as an ablation demonstrating why the fused design is justified.

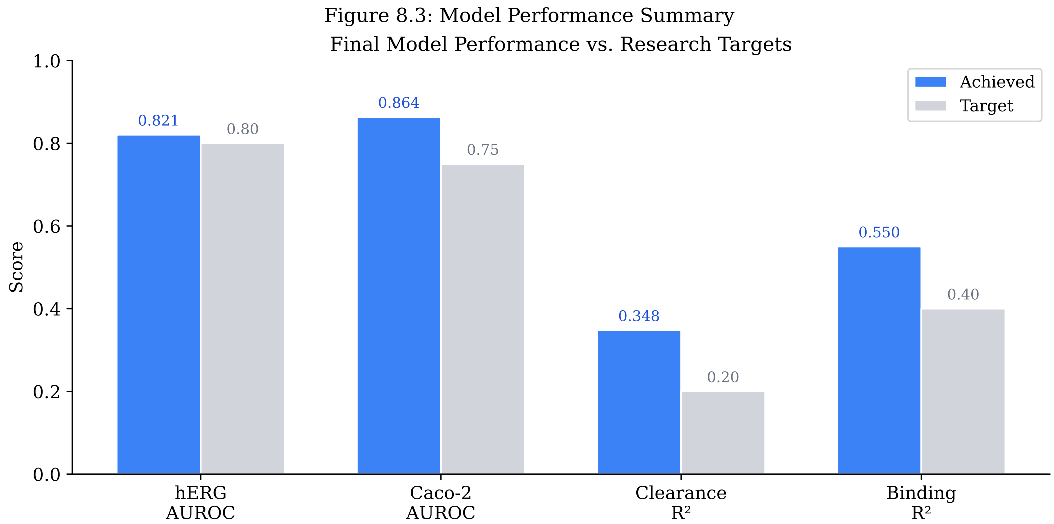

Figure 7.2: Summary of final predictive performance against the research targets. The upgraded predictive pipeline meets all four project targets, with the strongest binding performance achieved by the final fusion architecture.

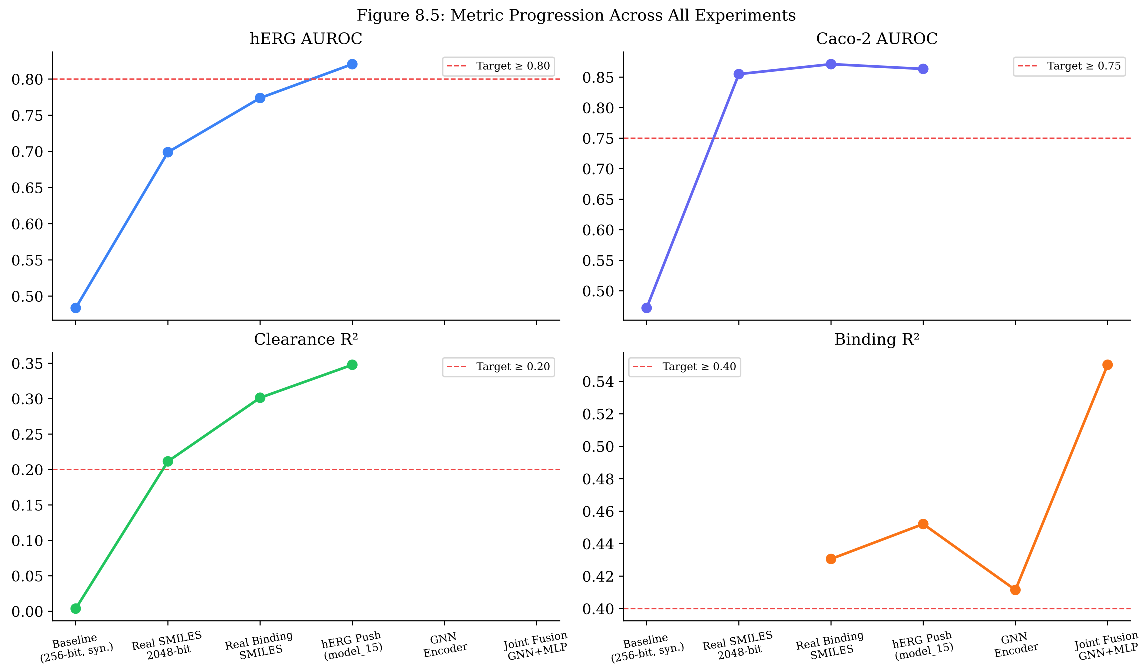

Figure 7.3: Progression of key metrics across major experimental milestones. The results show that meaningful gains arise first from the notebook-aligned TDC representation improvement, then from later real binding integration, and are finally consolidated by architectural refinement.

7.6. Neural ODE PK-PD Integration

The predictive modeling pipeline was next extended with a Neural ODE-based PK-PD module. This step transforms static molecular endpoint prediction into dynamic pharmacological simulation by mapping predicted clearance, permeability, binding affinity, and safety probabilities to PK-PD parameters and then solving the corresponding concentration-time and effect-time equations. The resulting simulations produce coherent PK decay profiles and interpretable pharmacodynamic saturation behavior.

This is a central result of the thesis because it links machine learning outputs to mechanistic downstream behavior. Compounds with higher predicted clearance exhibit faster concentration decay, whereas compounds with stronger predicted affinity achieve pharmacodynamic effect at lower concentrations through the inferred EC_{50} relationship. The framework therefore moves beyond isolated endpoint prediction and provides a mechanistically interpretable bridge between molecular structure and dynamic therapeutic response.

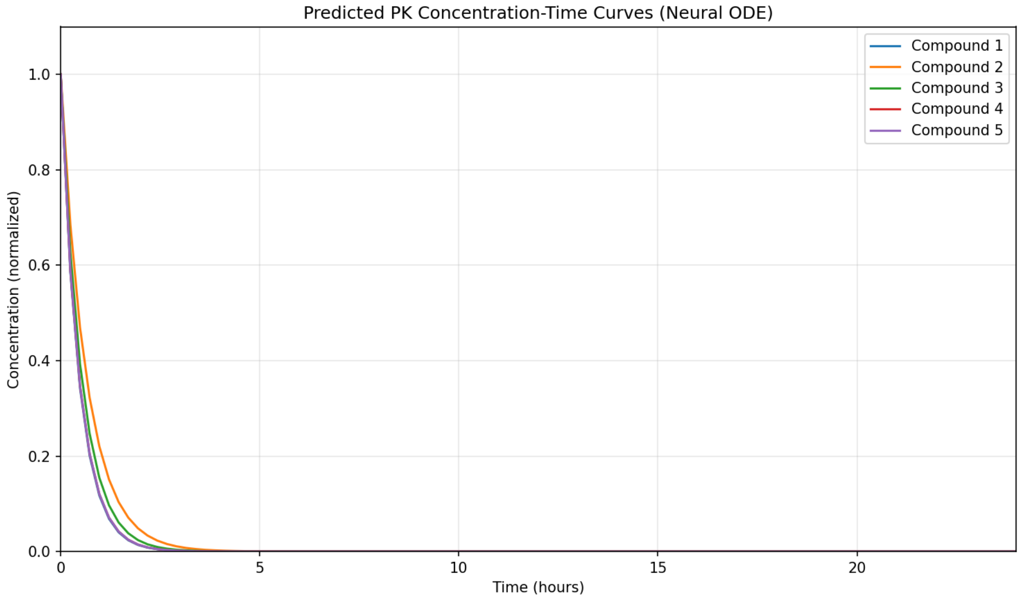

Figure 7.4: Representative Neural ODE PK-PD profiles. The concentration-time curves reflect predicted elimination behavior, while the effect-time curves show expected Emax-type pharmacodynamic saturation.

7.7. Probability Calibration and Robustness

Because later decision-oriented analyses depend on predicted probabilities rather than on ranking alone, post-hoc calibration was applied to the hERG and Caco-2 classifiers. Temperature scaling and Platt scaling both reduced expected calibration error, with temperature scaling yielding the most consistent improvement. This confirms that the raw classifier outputs were overconfident and that calibration was necessary before using these probabilities in safety-aware simulation and optimization.

Calibration is not the main novelty of the thesis, but it is essential for methodological credibility. Downstream decision support depends on whether predicted probabilities can be interpreted as plausible empirical risks. The observed ECE reductions therefore justify the use of calibrated hERG and Caco-2 probabilities in later scenario, optimization, and trial-simulation sections.

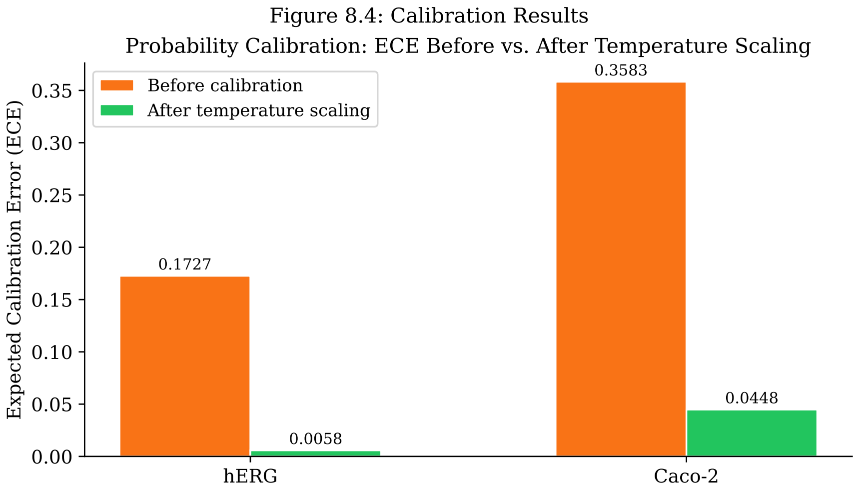

Figure 7.5: Expected Calibration Error (ECE) before and after post-hoc calibration. Calibration substantially improves the reliability of predicted probabilities for both safety-relevant classification tasks.

7.8. Pareto Efficacy-Safety Trade-Off Analysis

After integrating prediction and simulation, the framework was used to analyze efficacy-safety trade-offs across candidate regimens. Rather than optimizing efficacy alone, the analysis considered exposure-derived efficacy together with predicted hERG and Caco-2 risk. Pareto frontier analysis showed that no tested regimen simultaneously satisfied all efficacy and safety criteria, indicating that the problem is constrained by a genuine multi-objective trade-off rather than by incomplete optimization.

Within the set of Pareto-optimal regimens, a knee-point solution provides the most balanced trade-off between therapeutic effect and safety burden. This result is important because it demonstrates that the framework does not merely select the highest-efficacy regimen, but instead exposes the geometry of the underlying trade-off space. In this way, the model becomes a decision-support tool rather than a pure predictive engine.

Metric	**Knee-point regimen**
Dose	114.3 mg
CV IIV	0.150
Median efficacy exposure (AUC_E)	18.13
hERG event rate	83.10%
Caco-2 event rate	82.80%
Composite score	-9.2066
Distance to utopia	0.9907
Pareto feasibility	Not fully feasible under all constraints

Table 7.2: Selected knee-point regimen from the Pareto frontier analysis. Although no regimen satisfied all efficacy and safety constraints simultaneously, the knee-point provided the most balanced trade-off among the non-dominated solutions.

7.9. Virtual Clinical Trial Simulation

To assess translational utility, the selected regimen space was evaluated in a virtual clinical trial setting. Simulated treatment arms exhibited clear dose-response separation in efficacy, and the operating-characteristics analysis suggested very high statistical power under the modeled assumptions. At the same time, the simulated safety profile remained poor across active arms, consistent with the Pareto analysis and demonstrating that efficacy gains do not eliminate toxicity-related constraints.

This result should be interpreted as a model-based planning experiment rather than a literal prediction of real clinical trial outcomes. Its value lies in showing that molecular prediction, mechanistic simulation, and trial-like decision metrics can be linked within a single computational framework. In the thesis context, the virtual trial is therefore the final translational demonstration of the pipeline rather than a substitute for empirical validation.

Arm	**Dose (mg)**	**Median** AUC_E	**Median** $C_{\max}$	ΔAUC_E **vs placebo**	**Power**	**Mean hERG**
Placebo	0.0	0.00	0.00	—	—	0.0%
Low dose	68.6	13.54	6.76	+13.43	100.0%	98.3%
Knee-point	114.3	17.02	11.59	+16.91	100.0%	99.7%
High dose	160.0	19.48	16.13	+19.37	100.0%	99.9%

Table 7.3: Summary of the virtual clinical trial arms. Active doses produced strong efficacy separation from placebo and uniformly high simulated power, but the associated hERG burden remained high across all treatment arms.

7.10. Exploratory Extensions

An exploratory RapidDock-inspired branch was also implemented as a proof-of-feasibility study for geometry-aware protein-ligand modeling. The branch confirmed that distance-biased attention, symmetry-aware loss, and coordinate reconstruction can be trained end to end. However, due to the synthetic protein context and limited scale, the prototype is treated as a forward-looking research direction rather than as a production alternative to the main predictive pipeline.

7.11. Summary of Key Findings

The results support a clear staged interpretation of the framework. First, baseline performance is limited until structurally meaningful molecular inputs and real endpoint-aligned data are introduced. Second, real binding-data integration and hERG-focused refinement establish the predictive backbone that is later used for decision-oriented analysis. Third, post-hoc calibration is necessary to make the safety probabilities usable as risk quantities rather than only as ranking scores. Fourth, the joint GNN+MLP model provides the strongest final predictive architecture for binding-aware downstream modeling. Fifth, Neural ODE integration connects these predictive outputs to mechanistic PK-PD behavior in a coherent way. Finally, Pareto analysis, Monte Carlo uncertainty quantification, and virtual clinical trial simulation show that the framework can support translational decision analysis while also revealing that safety liabilities remain a dominant constraint even when predictive efficacy improves.

Taken together, these findings support the central thesis of this work: combining data-driven molecular prediction with mechanistic PK-PD simulation produces a more informative and translationally useful framework than either component alone. At the same time, the results also show that translational modeling must remain explicitly multi-objective, because higher predicted efficacy does not automatically imply clinically acceptable safety.

8. Discussion

8.1. Interpretation of Key Findings

The discussion chapter interprets the proposed neural PK-PD framework as a staged progression from molecular prediction to translational decision support. The central conclusion is not simply that one model variant performed best, but that useful pharmacological decision support emerged only after combining a transitional structural representation improvement, later real binding-data integration, hERG-focused refinement, probability calibration, fusion modeling, mechanistic simulation, and decision-oriented downstream analysis. The discussion therefore follows the same thesis-essential sequence as the results chapter rather than the broader exploratory notebook chronology.

8.1.1. Structural Representation Is a Necessary Starting Point

The first decisive result of the study is that predictive quality depends critically on the quality of the molecular representation. The transition from synthetic or structurally uninformative fingerprints to 2048-bit fingerprints derived from real TDC SMILES where available produced a substantial improvement across the main endpoints, especially for the classification tasks, even though the exploratory ChEMBL binding subset still remained a zero-fingerprint fallback at this stage. This finding shows that the early limitations of the baseline model were not solely architectural; they were strongly constrained by the absence of chemically meaningful structural information.

This result is significant because it establishes representation quality as a prerequisite for all later modeling steps. The poor baseline binding performance demonstrates that physicochemical descriptors alone are insufficient for learning robust structure-activity relationships in this setting. From a drug-discovery perspective, this reinforces a broader methodological point: complex models cannot compensate for structurally weak inputs. Consequently, the fingerprint upgrade was not a minor preprocessing refinement, but a foundational change that enabled the rest of the thesis.

8.1.2. Real Binding Data Provides the First Major Task-Specific Gain

The second major finding is that real binding information produces a task-specific gain beyond the general fingerprint upgrade. The introduction of real ChEMBL-derived binding data yielded the clearest improvement in affinity prediction, far stronger than would be expected from a generic feature-space expansion alone. This indicates that endpoint-aligned realism in the underlying data is essential, particularly for tasks such as binding affinity that depend directly on molecular recognition mechanisms.

This observation also clarifies the behavior of the multi-task framework. Although shared learning can transfer useful information across endpoints, each task still benefits most strongly from improvements in the data that are directly relevant to it. The larger gain in binding performance compared with the more modest secondary changes in the ADMET tasks therefore supports a disciplined interpretation of multi-task learning: shared representation is valuable, but it does not eliminate dependence on endpoint-specific data quality.

8.1.3. Predictive Refinement and Calibration Must Precede Translation

The architectural experiments show that stronger prediction required more than better features alone. The hERG-focused refinement demonstrated that model capacity and task-specific optimization choices, such as positive-class weighting, influence discrimination quality in meaningful ways. More importantly, the graph-based and fusion experiments showed that the best final architecture was not the most specialized single branch, but the joint GNN+MLP fusion model that combined graph-derived structural information with fingerprint-derived latent features.

However, predictive discrimination alone was not sufficient for translational use. The calibration experiments showed that even models with strong AUROC could still produce poorly calibrated probabilities. This distinction matters because the later PK-PD and regimen analyses interpret predicted probabilities as risk-related quantities rather than as ranking scores alone. The thesis therefore supports a broader methodological point: in decision-oriented modeling, discrimination and calibration are complementary properties, and both are required if model outputs are to be used in downstream pharmacological reasoning.

8.1.4. Fusion Modeling Benefits from Complementary Structural Views

The fusion results indicate that graph-derived and fingerprint-derived representations encode overlapping but non-identical information. Fingerprints efficiently summarize local substructures and chemically interpretable bit patterns, whereas graph neural networks learn adaptive topological features through message passing. The superior performance of the fused model therefore suggests that neither representation is complete on its own for the binding prediction problem addressed here.

This result is particularly important for the scientific positioning of the thesis. It shows that the work does not argue for replacing traditional molecular descriptors with graph learning, nor for rejecting graph-based models in favor of handcrafted fingerprints. Instead, the evidence supports a hybrid view in which each representation captures different aspects of molecular structure and their combination produces a stronger predictive basis for downstream pharmacological analysis.

8.2. Mechanistic PK-PD Interpretation

The Neural ODE simulation successfully translated predicted molecular properties into dynamic concentration-time and effect-time profiles with interpretable pharmacological behavior. This is one of the defining contributions of the thesis, because it shifts the framework from static endpoint prediction to mechanistic trajectory generation. The observed PK and PD trends are qualitatively consistent with expected behavior: higher predicted clearance leads to faster decline in concentration, while stronger predicted affinity produces higher effect at comparable exposure levels.

The value of this result lies not only in generating plausible curves, but in establishing a functional bridge between molecular prediction and dynamic therapeutic interpretation. In many drug-discovery workflows, molecular property prediction and PK-PD modeling remain separate analytical steps. By connecting them directly, the present framework supports a more coherent pipeline in which predicted structure-property relationships can be interpreted in terms of simulated exposure, effect, and downstream safety trade-offs.

8.3. Pareto, Monte Carlo, and Virtual Trial Implications

The Pareto analysis, Monte Carlo validation, and virtual clinical trial simulation collectively demonstrate that predictive improvement does not eliminate the central efficacy-safety tension. Across these downstream analyses, higher efficacy was repeatedly associated with persistently high safety burden, especially in the hERG-related dimension. This is not a failure of the framework; rather, it is an informative result revealing that the modeled regimen space is constrained by genuine competing objectives.

This finding has two implications. First, it validates the decision-oriented design of the framework, because a single-objective optimization strategy would have hidden this tension behind a superficially optimal regimen. Second, it suggests that translational utility in this domain depends not on finding one universally optimal solution, but on characterizing the trade-off landscape in a transparent manner. The knee-point regimen and virtual trial analyses are therefore best interpreted as tools for rational prioritization under constraint rather than as definitive clinical recommendations.

8.4. Limitations

1. **Public data quality**: All datasets are derived from public databases (ChEMBL, TDC, ToxCast, PK-DB). Measurement heterogeneity (different assay protocols, cell lines, species) introduces label noise that may limit maximum achievable performance.

2. **SMILES availability**: in the current notebook-aligned representation-upgrade stage, real SMILES are available for the TDC-derived tasks, but not for the exploratory ChEMBL binding snapshot. As a result, the ChEMBL rows retain zero-fingerprint fallback until the later real-binding integration step.

3. **ODE parameter assumptions**: The two-compartment PK model uses default population-average transfer constants (k_{12}, k_{21}) not predicted per-compound. Individualized PK parameter prediction would require PK-DB or clinical trial training data, which is limited in scale.

4. **GNN training data**: The GNN branch is trained only on binding affinity; the other three tasks retain MLP-only heads in the joint fusion model. Extension to multi-task GNN with graph inputs for all tasks is deferred to future work.

5. **Decision-layer assumptions**: The regimen optimization and virtual trial analyses rely on modeled efficacy and safety relationships rather than prospectively validated clinical outcomes. These sections should therefore be interpreted as translational feasibility studies, not as direct clinical prediction.

6. **RapidDock scope**: The RapidDock prototype is exploratory and not fully integrated with calibration, uncertainty quantification, or real protein structural context.

8.5. Comparison with Related Work

The multi-task predictive components of the framework are consistent with established ADMET modeling practice, where shared encoders and task-specific heads are commonly used to exploit correlations across endpoints. The graph-based and fusion components also align with recent molecular machine learning literature showing that graph encoders and fingerprint-based descriptors can be complementary rather than mutually exclusive.

The most distinctive aspect of the present work is the integration of these predictive models with a Neural ODE-based PK-PD simulation layer and subsequent decision-oriented analyses. While PK prediction, ADMET classification, and molecular graph learning are each well represented in the literature individually, relatively few studies connect them into a single pipeline that proceeds from molecular input to calibrated endpoint prediction, mechanistic simulation, efficacy-safety trade-off analysis, and virtual trial interpretation. In this sense, the contribution of the thesis is primarily integrative and translational rather than purely architectural.

9. Conclusion and Future Work

This thesis presents a complete neural PK-PD framework that links molecular prediction, calibration, mechanistic simulation, and decision analysis in one reproducible workflow. The combined approach improves endpoint modeling and enables clinically motivated in-silico analyses.

Future work should focus on:

1. external and prospective validation,
2. stronger mechanistic priors and uncertainty quantification,
3. expanded structure-aware modeling (including the RapidDock branch),
4. and deployment as an interactive decision-support tool.

Appendix

A.1 Data Acquisition Supplementary Tables

A.1.1 TDC ADMET Benchmark Datasets

Table A.1 details the eight TDC ADMET benchmark datasets used in the project. Each dataset provides curated molecular structures (SMILES) paired with standardized endpoint measurements from pharmaceutical literature.

Dataset Name	Category	Task Type	Source Publication
caco2_wang	Absorption	Classification	Wang et al., 2016
hia_hou	Absorption	Classification	Hou et al., 2004
solubility_aqsoldb	Absorption	Regression	AQSODB Database
lipophilicity_astrazeneca	Distribution	Regression	AstraZeneca Internal
ppbr_az	Distribution	Regression	AstraZeneca Internal
clearance_hepatocyte_az	Metabolism	Regression	AstraZeneca In Vitro
half_life_obach	Metabolism	Regression	Obach et al., 2008
herg	Toxicity	Classification	Comprehensive (Gaulton et al.)

Table 9.1: A.1: TDC ADMET Benchmark Datasets–Categories, Task Types, and Sources.

A.1.2 ChEMBL Real-SMILES Targets

Table A.2 lists the eight pharmacologically important targets from which real SMILES binding data was extracted from ChEMBL. These targets span major pharmaceutical categories: G-protein coupled receptors (GPCRs), kinases, and nuclear receptors.

A.1.3 PK Parameters Extracted from PK-DB

Table A.3 lists the PK parameters available in the PK-DB database. These parameters are essential for population PK modeling and ODE-based simulation in downstream analysis.

Target ID	**Target Name**	**Target Class**	**Therapeutic Area**
CHEMBL217	Dopamine D2 receptor	GPCR	CNS; antipsychotics, Parkinson's
CHEMBL251	Adenosine A2a receptor	GPCR	Neuroinflammation; caffeine antagonism
CHEMBL203	EGFR	Kinase	Oncology; lung cancer
CHEMBL301	CDK2	Kinase	Cell cycle; oncology
CHEMBL210	Beta-2 adrenergic receptor	GPCR	Respiratory; asthma, COPD
CHEMBL1871	Androgen receptor	Nuclear Receptor	Prostate cancer; hormone therapy
CHEMBL2034	Glucocorticoid receptor	Nuclear Receptor	Inflammation; immunosuppression
CHEMBL325	Serotonin 5-HT2A receptor	GPCR	Depression; psychedelics

Table 9.2: A.2: ChEMBL real-SMILES targets used in the thesis.

Parameter	**Description**
CL	Clearance (total body, mL/min or L/h)
Vd	Volume of distribution (L)
AUC	Area under the plasma concentration-time curve
Cmax	Maximum (peak) plasma concentration
Tmax	Time to reach peak concentration
$t_{1/2}$	Elimination half-life (hours)
ka	Absorption rate constant (oral, h^{-1})
F	Bioavailability (fraction of dose absorbed systemically)
MRT	Mean residence time
λ_z	Elimination rate constant (h^{-1})
fu	Fraction unbound (free drug in plasma)
PPB	Plasma protein binding (percent bound)

Table 9.3: A.3: PK Parameters Extracted from PK-DB.

A.1.4 Data File Inventory and Sizes

Table A.4 provides a complete inventory of all raw data files produced by the download pipeline, organized by source directory, with approximate file sizes in kilobytes.

Directory	File	Size (KB)
pubchem/	assay_herg_qhts_aid588834.csv	150
pubchem/	assay_cyp3a4_inhibition_aid54772.csv	180
pubchem/	compound_warfarin.sdf	50
pubchem/	compound_midazolam.sdf	55
pubchem/	compound_caffeine.sdf	40
pkdb/	studies.json	2,500
pkdb/	studies_top100.json	250
pkdb/	pkdb_studies_complete.json	5,000
pkdb/	pkdb_studies_summary.csv	200
tdc/	tdc_caco2_wang.csv	150
tdc/	tdc_hia_hou.csv	120
tdc/	tdc_solubility_aqsoldb.csv	180
tdc/	tdc_lipophilicity_astrazeneca.csv	140
tdc/	tdc_ppbr_az.csv	130
tdc/	tdc_clearance_hepatocyte_az.csv	110
tdc/	tdc_half_life_obach.csv	95
tdc/	tdc_herg.csv	160
tdc/	tdc_metadata.json	25
chembl/	chembl_binding_affinity.csv	300
chembl/	chembl_binding_affinity_metadata.json	15
chembl/	chembl_binding_affinity_with_smiles.csv	450
chembl/	chembl_binding_affinity_with_smiles_metadata.json	20
toxcast/	toxcast_representative.csv	8,000
toxcast/	toxcast_high_priority.csv	4,500
toxcast/	toxcast_critical_priority.csv	2,000
toxcast/	toxcast_representative_metadata.json	50
	TOTAL	**∼25,000+**

Table 9.4: A.4: Complete Raw Data File Inventory and Approximate Sizes.

A.1.5 Data Download Pipeline Execution Timeline

The consolidated data download pipeline automatically logs execution time for each section (cell). Table A.5 illustrates representative timings for a full pipeline run (times vary based on network conditions and API responsiveness).

Section	Description	Data Source	Typical Duration
§1A	Imports	Python libraries	2–5 s
§1B	Configuration	API endpoints, paths	1 s
§1C	Utility functions	Download helpers	1 s
§2	PubChem downloads	PubChem REST API	15–30 s
§3A	PK-DB functions	PK-DB API setup	5 s
§3B	PK-DB complete fetch	PK-DB bulk API	30–60 s
§3C	PK-DB summary	Aggregation	10 s
§4	TDC ADMET datasets	PyTDC client	20–40 s
§5A	ChEMBL synthetic	Synthetic generation	5 s
§5B	ChEMBL real SMILES	ChEMBL REST API	120–180 s
§6	ToxCast generation	Synthetic generation	10 s
§7	Summary/Verification	File checks	5 s
§8	Timeline reporting	Aggregation	2 s
	Total (all sections)		**∼5–10 min**

Table 9.5: A.5: Data Download Pipeline Section: Typical Execution Times.

A.2 Data Quality Assurance Checklist

Pre-Processing QA Steps

The following checklist summarizes QA procedures applied before model training:

1. **SMILES Validation**: All SMILES strings pass RDKit canonicalization. Invalid SMILES are excluded from analysis.

2. **Molecular Descriptor Computation**: MW, logP, TPSA, HBA, HBD, RotBonds computed via RDKit from canonical SMILES.

3. **Missing Data Handling**: NaN/Inf values identified and reported. Imputation strategy (mean, median, or exclusion) applied per task.

4. **Duplicate Detection**: Exact-match and SMILES-based duplicates identified. For binding data, activities averaged; for classification, consensus labels computed.

5. **Train/Val/Test Leakage Check**: No molecule (by SMILES) appears in multiple splits. Scaffold-based similarity checks performed.

6. **Class Balance Assessment**: For classification tasks, label distributions reported and imbalance metrics documented.

7. **Outlier Flagging**: Molecules with extreme property values (e.g., MW > 600 g/mol) flagged but retained (not removed).
8. **Provenance Documentation**: Metadata files record source, version, download date, and license for each dataset.

A.3 Reproducibility Notes

Random Seed Configuration

To ensure reproducible data splits and model initialization:

```
import numpy as np
import torch
np.random.seed(42)
torch.manual_seed(42)
torch.cuda.manual_seed_all(42)
```

All split generation uses `random_state=42` in scikit-learn functions.

Required Python Packages

Core dependencies for data download and preprocessing:

```
numpy>=1.19.0
pandas>=1.2.0
requests>=2.25.0
rdkit>=2021.03.1
chembl-webresource-client>=0.10.9
pytdc>=0.4.0
PyYAML>=5.4.0
torch>=1.9.0
scikit-learn>=0.24.0
```

Hardware and Runtime Notes

Typical environment for full pipeline execution:

1. **CPU**: Multi-core processor (4+ cores recommended for parallel RDKit operations).

2. **RAM**: 8 GB minimum, 16 GB recommended for full dataset loading.

3. **Storage**: 30+ GB free space for raw data directory (`data/raw/`).

4. **Network**: Stable internet connection required for API downloads. Pipeline includes automatic retry and resume capability.

5. **Runtime**: Full pipeline (all sources, first-time download) typically 5–10 minutes; reuse of cached files (skip download) takes <1 minute.

Notebook Artifact Contract

The experimental notebooks communicate through persisted phase artifacts rather than ad hoc in-memory transfer:

- The feature-engineering notebook writes feature matrices, scalers, and metadata to `data/processed/phase3a\_artifacts/`.
- The model-design notebook loads those artifacts, augments the binding task with the RapidDock-derived bridge feature, and writes model state, loader state, history, and raw arrays to `data/processed/phase3b\_artifacts/`.
- The fine-tuning notebook reloads the model-design baseline, performs fine-tuning and threshold selection, and writes production-ready artifacts to `data/processed/phase3c\_artifacts/`.
- The experiment notebook loads the fine-tuning production bundle and writes figures and summary tables to `data/raw/` and `data/outputs/`.

In the current implementation, these notebook stages discover the nearest valid `data/` root dynamically. This makes the execution flow less sensitive to notebook relocation within the project while preserving the same artifact directories described in the thesis.

A.4 Hyperparameter Configuration

Table A.6 summarises the fixed and searched hyperparameters used across Phase 3 model training.

Parameter	**Value / Search Range**	**Selected**
Optimizer	Adam	Adam
Learning rate	10^{-3}	10^{-3}
Weight decay (L2)	10^{-4}	10^{-4}
Gradient clip (max norm)	1.0	1.0
Batch size	64	64
Max epochs	300	300
Early stopping patience	40 (60 for hERG push)	60
LR scheduler	ReduceLROnPlateau	factor=0.5, patience=10
hidden_dim	{128, 256}	256 (model_15)
pos_weight (hERG)	{0.5, 1.0, 1.5, 2.0, 3.0}	1.5
Fingerprint bits	{256, 2048}	2048 (Morgan, radius=2)
Dropout (encoder)	0.3	0.3
Dropout (MLP head)	0.1	0.1
Focal loss γ	2.0	2.0

Table 9.6: A.6: Hyperparameter configuration summary. Search ranges show grid values evaluated; Selected shows the value used in the final production model (model_15, 623,078 parameters).

A.5 Extended Results by Experiment

Table A.7 provides the full progression of test-set metrics across all major experimental milestones from the baseline (synthetic fingerprints) through to the final fusion model.

Calibration results (post-hoc, validation set):

- hERG ECE before calibration: 0.1727; after temperature scaling: **0.0058**.
- Caco-2 ECE before calibration: 0.3583; after Platt scaling: **0.0448**.

Experiment	hERG AUROC	Caco-2 AUROC	Clearance R^2	Binding R^2
Baseline (synthetic FP, 256-bit)	0.4836	0.4719	0.0037	0.0019
Real SMILES, 2048-bit FP	0.6989	0.8550	0.2115	–
Real binding SMILES (ChEMBL)	0.7738	0.8713	0.3013	0.4306
hERG push (hidden_dim=256, pos_weight=1.5)	**0.8206**	0.8635	0.3478	0.4521
GNN encoder (MPNN, 3-layer)	–	–	–	0.4115
Joint GNN+MLP fusion	–	–	–	**0.5502**

Table 9.7: A.7: Test-set metric progression across the main experimental milestones. Dashes indicate metrics outside the primary focus of that experiment.

A.6 Software Environment and Version Control

The project is distributed as a Python notebook environment with:

1. **Python version**: 3.14.2 in the validated `venv_pkpd` environment
2. **Package manager**: pip (via venv or conda)
3. **Source control**: Git (see project repository for version history)
4. **Development environment**: Jupyter Notebook interfaces, VS Code
5. **Reproducibility framework**: Fixed seeds, version pinning, and execution logging

This appendix provides the detailed reference material necessary for external validation and reproducibility of the data acquisition and preprocessing pipeline.

A.7 Technical Specifications: APIs, Configuration, and Download Settings

A.7.1 API Endpoints and Request Parameters

PubChem REST API:

`Base: https://pubchem.ncbi.nlm.nih.gov/rest/pug`

```
  Assay:    /assay/aid/{AID}/CSV
  Compound: /compound/name/{NAME}/SDF?record_type=3d
```

PK-DB REST API v1:

```
Base: https://pk-db.com/api/v1
  Studies: /studies/?format=json
```

ChEMBL REST API:

```
Base: https://www.ebi.ac.uk/chembl/api/data
  Activity: /activity/?target_chembl_id={ID}&
            standard_type__in=IC50,Ki,EC50&
            pchembl_value__isnull=False&
            limit=1000&offset={offset}
```

TDC (Python SDK, not REST):

```
from pytdc import admet_group
data = admet_group().get(dataset_name)
```

A.7.2 Download Configuration Parameters

The pipeline uses the following runtime configuration to ensure robustness and courtesy:

1. **Request timeout**: 60 seconds for most APIs, 120 seconds for bulk PK-DB queries.
2. **ChEMBL rate-limit delay**: 0.35 seconds between consecutive API page requests (respects EBI service terms).
3. **ChEMBL page size**: 1,000 records per API request.
4. **ChEMBL cap per target**: 600 records maximum to ensure diversity.
5. **Retry logic**: 3 attempts per failed API call with 1.5 second exponential backoff between retries.

A.7.3 File System Organization

All raw data is organized under `data/raw/` with per-source subdirectories:

```
data/raw/
|-- pubchem/
```

```
|   |-- assay_herg_qhts_aid588834.csv
|   |-- assay_cyp3a4_inhibition_aid54772.csv
|   `-- compound_{warfarin,midazolam,caffeine}.sdf
|-- pkdb/
|   |-- studies.json
|   |-- studies_top100.json
|   |-- pkdb_studies_complete.json
|   `-- pkdb_studies_summary.csv
|-- tdc/
|   |-- tdc_caco2_wang.csv  ...
|   `-- tdc_metadata.json
|-- chembl/
|   |-- chembl_binding_affinity.csv (synthetic)
|   `-- chembl_binding_affinity_with_smiles.csv (real)
`-- toxcast/
    |-- toxcast_representative.csv
    |-- toxcast_high_priority.csv
    `-- toxcast_critical_priority.csv
```

List of symbols and abbreviations

ADME	Absorption, Distribution, Metabolism, Excretion
PK/PD	Pharmacokinetics / Pharmacodynamics
ODE	Ordinary Differential Equation
GNN	Graph Neural Network
MPNN	Message Passing Neural Network
AUROC	Area Under Receiver Operating Characteristic
ECE	Expected Calibration Error
RMSE	Root Mean Squared Error
IIV	Inter-individual Variability

List of Figures

4.1 ChEMBL Binding Affinity Distribution and Target Coverage. Left panel: histogram of pIC50 values (converted from IC50 [nM]) across 2,000 compounds, mean pIC50 = 6.09, with mean overlay (red dashed line). Right panel: top 10 pharmacological targets by compound frequency (kinases, GPCRs, nuclear receptors). 28

4.2 Illustrative toxicity risk stratification in the aggregated ToxCast-style reference set (570,167 rows across critical-, high-, and representative-priority files). Left panel: compound counts across priority tiers. Right panel: activity rates across the most prevalent toxicity assay endpoints. The figure summarizes broad safety-screening structure rather than defining a final supervised training target. 29

4.3 Exploratory benchmark distributions for the three TDC endpoints most relevant to the PK-PD framework. Left: hERG class balance in the benchmark subset used for visualization. Middle: Caco-2 permeability distribution. Right: hepatocyte-clearance distribution showing heterogeneous metabolic stability. These panels illustrate endpoint behavior prior to final harmonized dataset assembly. 31

7.1 Training and validation loss curves for the baseline multi-task model. The smooth decrease in both curves confirms stable optimization of the shared encoder and task-specific heads, including the differentiable PK component. 65

7.2 Summary of final predictive performance against the research targets. The upgraded predictive pipeline meets all four project targets, with the strongest binding performance achieved by the final fusion architecture. 67

7.3 Progression of key metrics across major experimental milestones. The results show that meaningful gains arise first from the notebook-aligned TDC representation improvement, then from later real binding integration, and are finally consolidated by architectural refinement. 68

7.4 Representative Neural ODE PK-PD profiles. The concentration-time curves reflect predicted elimination behavior, while the effect-time curves show expected Emax-type pharmacodynamic saturation. 69
7.5 Expected Calibration Error (ECE) before and after post-hoc calibration. Calibration substantially improves the reliability of predicted probabilities for both safety-relevant classification tasks. 70

List of Tables

1 Organization of the thesis. 9

1.1 Mapping of predicted molecular properties to ODE pharmacokinetic/pharmacodynamic parameters. 11

1.2 Expected Calibration Error before and after post-hoc probability calibration. . . 13

2.1 Summary of the main literature strands relevant to the thesis and the gap addressed by their incomplete integration. 17

3.1 Mapping of multi-task model outputs to Neural ODE pharmacokinetic/pharmacodynamic parameters. 19

3.2 Research performance targets and final achieved metrics (model_15 / joint GNN+MLP fusion). 21

4.1 Representative toxicity categories used to illustrate multi-domain safety screening in a ToxCast/Tox21-style reference dataset. 25

4.2 Scientific role of each data source in the thesis framework. 26

4.3 Source-level overview used during exploratory analysis. Counts refer to the specific benchmark subsets or aggregated reference files inspected in this section and are distinct from the later harmonized task counts used for model training. . . . 27

4.4 ToxCast Risk Level Distribution across 332,507 Assay Results. 29

4.5 Baseline multi-task learning setup for the harmonized feature-engineering matrix. Later chapters report upgraded dataset variants. 35

4.6 Feature-engineering notebook hyperparameter configuration for the baseline harmonized matrix. Binding and Caco-2 receive elevated loss weights to stabilize optimization of the harder and less homogeneous tasks in multi-task training. . . 37

5.1 Model-design notebook target metrics. The architecture is evaluated against these thresholds to determine readiness for the fine-tuning notebook. 45

5.2 Representation-upgrade comparison aligned with the current executed notebook. The upgrade increases fingerprint resolution to 2,048 bits and introduces real structural signal for the TDC-derived tasks, while the exploratory ChEMBL binding subset remains a zero-fingerprint fallback because SMILES are not available in the current snapshot. 51
5.3 Mapping of predicted molecular properties to ODE pharmacokinetic/pharmacodynamic parameters. 54

6.1 Revised Phase 3D thesis-essential experimental sequence. The main narrative emphasizes the experiments that most directly improved predictive quality, calibration, mechanistic interpretability, and translational decision support. Intermediate exploratory analyses are discussed only as supporting evidence. 62

7.1 Summary of the molecular representation upgrade aligned with the current notebook state. The 2,048-bit encoding introduces real structural signal for the TDC-derived tasks while the exploratory ChEMBL subset remains a zero-fingerprint fallback. 66
7.2 Selected knee-point regimen from the Pareto frontier analysis. Although no regimen satisfied all efficacy and safety constraints simultaneously, the knee-point provided the most balanced trade-off among the non-dominated solutions. 71
7.3 Summary of the virtual clinical trial arms. Active doses produced strong efficacy separation from placebo and uniformly high simulated power, but the associated hERG burden remained high across all treatment arms. 71

9.1 A.1: TDC ADMET Benchmark Datasets–Categories, Task Types, and Sources. . 79
9.2 A.2: ChEMBL real-SMILES targets used in the thesis. 80
9.3 A.3: PK Parameters Extracted from PK-DB. 80
9.4 A.4: Complete Raw Data File Inventory and Approximate Sizes. 81
9.5 A.5: Data Download Pipeline Section: Typical Execution Times. 82
9.6 A.6: Hyperparameter configuration summary. Search ranges show grid values evaluated; Selected shows the value used in the final production model (model_15, 623,078 parameters). 85
9.7 A.7: Test-set metric progression across the main experimental milestones. Dashes indicate metrics outside the primary focus of that experiment. 86

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List of appendices

1. Hyperparameter search grids and training configuration snapshots.
2. Extended performance tables and calibration outputs.
3. Supplementary simulation plots (population, Pareto, and trial analyses).
4. Reproducibility sheet: package versions, seed settings, and runtime notes.